



**COMSATS University Islamabad
Abbottabad, Pakistan**

Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection

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Bachelor of Science in Software Engineering (2022-2026)

The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others.



COMSATS University, Islamabad Pakistan

Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection

**A project presented to
COMSATS University Islamabad, Abbottabad Campus**

**In partial fulfillment
Of the requirement for the degree of**

Bachelor of Science in Software Engineering (2022-2026)

By

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FINAL APPROVAL**

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EXECUTIVE SUMMARY

The Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection is an AI-assisted web application developed to support ECG-based cardiovascular-condition analysis. The system combines automated ECG-image processing, machine-learning prediction, explainability-oriented outputs, and a structured clinical workflow. It is designed to assist healthcare professionals and does not replace the judgement of qualified doctors.

The platform supports four users: Admin, Doctor, Technician, and Patient. Patients can register and access approved reports directly, while doctors and technicians require Admin approval before login. Technicians upload ECG images and assign cases, doctors review AI-generated findings and approve or reject reports, and patients can upload ECG and view approved results.

The system processes ECG-sheet images by validating the input, separating the twelve ECG leads, applying preprocessing and contour extraction, reducing features through PCA, and generating predictions through a soft-voting ensemble model. It identifies four categories: Normal ECG, Abnormal Heartbeat, Myocardial Infarction, and History of Myocardial Infarction. The platform also presents probabilities, model votes, annotated ECG images, and interpretation notes to support transparent clinical review.

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Ferdous Gulzar



Muhammad Asim Ilyas



ABBREVIATIONS

Abbreviation	Definition
SRS	Software Require Specification
PC	Personal Computer
App(s)	Application(s)
UI	User-Interface
UX	User-Experience
API	Application Programming Interface
SPM	Software Project Management
OOP	Object Oriented Programming
HCI	Human Computer Interaction
FR	Functional Requirement
NFR	Non-Functional Requirement
SDLC	Software Development Life Cycle
iOS	iPhone Operating System
MVC	Mobile View Controller
DBMS	Database Management System
CLI	Command-Line Interface

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1. Introduction

This chapter provides an overview of the project, outlining its purpose, objectives, and the methodologies adopted for development. It establishes the foundation for understanding the motivation behind creating the Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection and highlights the scope, technical significance, and real-world impact of the system.

1.1. Brief

This project focuses on developing a web-based AI-powered clinical decision support system titled “Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection.” The system is designed to analyze ECG signals and images to assist in early diagnosis of cardiovascular diseases. The outcome includes a feature-rich platform with functionalities such as ECG upload, automated preprocessing, AI-based prediction, explainable visualization, and report generation. Technologies such as React (TypeScript), Python (FastAPI) [1], TensorFlow, PyTorch, and Supabase [2] were used in the development process. The system follows a microservices-based architecture [3], ensuring scalability, modularity, and efficient data processing. This report presents all major phases of development, including requirement analysis, system design, implementation, testing, and future enhancements.

1.2. Relevance to Course Modules

The following course modules contributed significantly to the successful development of this project by providing the necessary theoretical knowledge and practical skills required for building an AI-driven healthcare system.

1.2.1. Software Engineering:

Core concepts of software engineering were applied, including requirement analysis, system design, iterative development, and testing. These ensured a structured and maintainable development process for the overall system..

1.2.2. Object Oriented Programming:

Object-oriented principles were used to design modular components such as user roles, ECG records, prediction models, and report generation modules, ensuring code reusability and system scalability.

1.2.3. Database Systems:

Database concepts were implemented using Supabase (PostgreSQL) [2] for managing user data, ECG records, predictions, and reports. Secure storage and efficient retrieval of medical data were key aspects of this module.

1.2.4. Intelligence / Machine Learning:

This module played a central role in the project by enabling ECG signal preprocessing, feature extraction, classification using multiple models, and hybrid fusion techniques

for improved diagnostic accuracy.

1.2.5. Human Computer Interaction:

HCI principles were applied to design intuitive dashboards and interfaces for doctors, patients, technicians, and admins, ensuring usability, accessibility, and efficient workflow interaction.

1.2.6. Web Development:

Since the system is a web-based platform, web engineering concepts were used to develop the frontend using React (TypeScript) and backend APIs using FastAPI. [1] This enabled seamless communication between client and server, ensuring efficient data handling and user interaction.

1.3. Project Background

The idea behind the Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection is to develop an intelligent and reliable system that assists healthcare professionals in diagnosing cardiovascular diseases using ECG data. Traditional ECG interpretation often requires expert analysis and can be time-consuming, which may delay diagnosis and treatment. This project addresses these challenges by integrating artificial intelligence with clinical workflows to provide faster and more accurate diagnostic support.

The system is trained and evaluated using the ECG Images Dataset of Cardiac Patients [4], a publicly available benchmark dataset comprising ECG images from patients with Myocardial Infarction, abnormal heartbeat, History of MI, and normal individuals, developed under the auspices of Ch. Pervaiz Elahi Institute of Cardiology, Multan, Pakistan. This enhances prediction accuracy and reliability while also providing explainable outputs to support clinical decision-making. The platform follows a microservices-based architecture [3], allowing efficient handling of ECG uploads, AI processing, and report generation in a scalable environment.

The primary objectives of this project are to develop a robust AI-based diagnostic platform capable of accurate ECG analysis, to implement hybrid fusion techniques for improved prediction performance, and to provide explainable AI outputs that enhance trust and transparency. Additionally, the system aims to deliver a secure, user-friendly interface for multiple stakeholders, including doctors, patients, technicians, and administrators, ensuring smooth integration into real-world healthcare workflows.

1.4. Literature Review

The rapid advancement of digital healthcare technologies has significantly transformed medical diagnosis systems, particularly with the integration of artificial intelligence in cardiovascular disease detection. In recent years, there has been a growing demand for intelligent systems capable of analyzing ECG signals efficiently and assisting clinicians in early diagnosis. While several existing solutions provide ECG analysis and prediction capabilities, many lack comprehensive features such as multimodal data processing, explainable AI, and integrated clinical workflows. The following Table 1.1 presents a

comparative analysis of existing systems and highlights their limitations along with the proposed solution.

Table 1.1: Literature Review/System Analysis

Application / System	Weakness	Proposed Project Solution
Basic ECG ML Models — Nawaz et al. (2024) [5]	Focus only on single-model predictions (DT, LR, RF, SVM) on static ECG images, resulting in limited accuracy and lack of robustness.	The Hybrid Multimodal Fusion Framework integrates multiple models and applies fusion techniques to improve prediction accuracy and reliability.
Existing ECG Analysis Tools — Acharya et al. (2017) [6]	Deep CNN model classifies heartbeats but lacks explainability, making predictions difficult for doctors to interpret (black-box model with no XAI support).	The proposed system incorporates Explainable AI (XAI) techniques using SHAP and LIME to provide visual and feature-based explanations for predictions.
Hospital ECG Systems — Rajpurkar et al. (2017) [7]	Demonstrates that traditional hospital ECG interpretation relies entirely on manual cardiologist analysis, which is time-consuming and limited in scalability across high patient loads.	The system provides automated ECG preprocessing, AI-based analysis, and report generation to support faster and more consistent clinical decision-making.
Research-based ECG Systems — Nayyab et al. (2025) [8]	CNN and DNN models focus only on prediction using the PTB-XL dataset with no role-based access, no clinical workflow support, and no structured reporting or user management.	The proposed platform includes role-based access (Admin, Doctor, Technician, Patient) and full workflow integration from ECG upload to AI prediction to report approval.

1.5. Analysis from Literature Review

The literature review highlights several critical challenges in existing ECG-based cardiovascular disease detection systems, particularly in terms of accuracy, interpretability, automation, and system integration. Many existing solutions rely on single-model approaches [5], which often result in limited predictive performance and reduced reliability across different ECG datasets. Furthermore, most systems provide results without clear explanations [6], making it difficult for healthcare professionals to fully trust and interpret the outcomes in clinical practice.

Another major limitation observed in existing systems is the lack of automation in the diagnostic workflow. In many healthcare environments, ECG interpretation is still performed manually by specialists, making the process time-consuming and potentially error-prone. Additionally, several research-based [8] systems focus only on prediction

functionality and do not provide complete system features such as user management, report handling, and structured workflows, which limits their real-world applicability.

In contrast, the Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection is designed to address these limitations by introducing a more comprehensive and integrated approach. The system processes ECG data through multiple models and combines their outputs using a hybrid fusion mechanism, improving overall prediction accuracy and consistency. This approach enhances reliability compared to single-model systems.

A key strength of the proposed system is the inclusion of explainability features, which provide meaningful insights into prediction results. This allows doctors to better understand the reasoning behind system outputs, increasing confidence in the diagnostic process. Additionally, the system automates the complete ECG analysis workflow, including upload, preprocessing, analysis, and report generation, reducing manual effort and improving efficiency.

Another significant improvement is the development of a fully integrated web-based platform with role-based access for Admin, Doctor, Technician, and Patient. This ensures secure data handling, organized workflows, and better coordination among users. The system is also designed with scalability in mind, making it suitable for real-world deployment in clinical environments.

In conclusion, the proposed system effectively addresses the limitations identified in existing solutions by improving accuracy, introducing explainability, enabling automation, and providing a complete end-to-end platform. It offers a more reliable and practical approach to cardiovascular disease detection compared to traditional and existing ECG analysis systems.

1.6. Methodology and Software Lifecycle for This Project

This project follows an Agile methodology combined with an iterative and incremental development process to ensure flexibility, continuous improvement, and efficient handling of complex system components. The selected Software Development Life Cycle (SDLC) model is Agile due to its adaptability, collaborative nature, and focus on delivering functional system modules in stages. This approach is particularly suitable for this project as it involves multiple components such as web development, AI-based ECG analysis, and system integration. It allows continuous feedback from users and stakeholders, enabling refinement of system features and improvement of overall performance throughout the development lifecycle. [9]

1.6.1. Rationale behind Selected Methodology

For this AI-based web application project, the choice of methodology focuses on managing both system development and analytical components effectively. The iterative approach offers several advantages that align with the dynamic nature of

developing a healthcare system involving ECG processing, multiple models, and fusion techniques. It allows gradual development and testing of individual modules such as authentication, ECG upload, preprocessing, and prediction, ensuring system stability and accuracy.

1.6.2. Flexibility

System requirements and performance expectations may evolve during development, especially when working with ECG data and analytical models. An iterative process provides flexibility to modify features, improve algorithms, and refine system components without disrupting the entire development flow.

1.6.3. Early Feedback

By delivering functional modules such as ECG upload, analysis, and report generation in each iteration, feedback can be collected from users and evaluators early in the process. This helps identify usability issues, improve system performance, and enhance the accuracy and reliability of results.

1.6.4. Rapid Prototyping

The iterative approach enables quick development of prototypes, such as dashboards and ECG analysis interfaces, allowing stakeholders to visualize system functionality early. This improves communication, reduces design errors, and ensures that the system aligns with user expectations.

1.6.5. Continuous Improvement

Each development cycle provides an opportunity to evaluate system performance and refine components. Improvements can be made in areas such as ECG preprocessing, prediction accuracy, user interface design, and system integration, leading to a more robust and reliable application.

1.6.6. Adaptability

Agile promotes collaboration among team members and stakeholders throughout the project lifecycle. Regular communication ensures proper coordination between development, system design, and analytical components, leading to a well-integrated solution.

1.6.7. Collaborative Approach

Agile promotes collaboration among team members and stakeholders throughout the project lifecycle. Regular communication ensures proper coordination between development, system design, and analytical components, leading to a well-integrated solution.

1.6.8. Incremental Delivery

The system is developed in increments, where each module such as authentication,

ECG upload, analysis, and reporting is completed and integrated step by step. This approach enables early deployment of functional features and allows continuous refinement based on feedback, ultimately improving the system

2. Problem Definition

This chapter explains the exact problem addressed by the project and defines the expected solution outcome. It focuses on the challenges in developing an accurate, efficient, and interactive web-based system for cardiovascular disease detection using ECG data and AI-based analysis. It also highlights the need for a reliable digital healthcare platform that supports medical staff in decision-making.

2.1. Problem Statement

Cardiovascular diseases are among the leading causes of death worldwide, and early detection plays a critical role in reducing mortality rates. However, traditional ECG interpretation methods rely heavily on manual analysis by medical experts, which can be time-consuming, error-prone, and dependent on individual experience. This creates a gap in providing fast, consistent, and scalable diagnostic support, especially in high-patient-load environments.

Another major challenge is the lack of integrated systems that combine ECG data processing, multiple model-based predictions, and explainable AI in a single platform. Existing systems often provide either basic signal viewing [5] or isolated predictive models without proper workflow integration for doctors, technicians, and patients [8]. This limits their practical use in real clinical environments.

The problem is to develop a Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection that can efficiently process ECG data (signals/images), apply multiple AI models for classification, and generate reliable diagnostic results with explainability support. The system must ensure secure role-based access for Admin, Doctor, Technician, and Patient, while also enabling smooth workflows for ECG upload, preprocessing, prediction, report generation, and approval.

The expected outcome of this project is a fully functional, web-based clinical decision support system that improves ECG analysis accuracy, reduces manual workload, and enhances diagnostic efficiency. The system should provide fast, reliable, and interpretable predictions using multiple models, along with structured reporting and a user-friendly interface to support better healthcare decision-making.

2.2. Deliverable and Development Requirements

This section defines the expected outputs of the project along with the core technical and system requirements needed for development. It serves as a structured guideline for what the system must deliver and the technologies, modules, and resources required to implement the Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection. It also provides a basis for evaluating the success and completeness of the project.

2.2.1. Deliverables

This section describes the final outcomes produced by the project, including the fully developed web application, supporting documentation, system reports, and AI-based diagnostic modules. It clearly defines what will be delivered at the completion of the project.

2.2.2. Web-Based Application:

A fully functional web-based system has been developed for cardiovascular disease detection. The application supports multiple user roles including Admin, Doctor, Technician, and Patient, and provides secure access through role-based authentication. It includes features such as ECG upload, AI-based prediction, report generation, and diagnostic review.

2.2.3. User Interface Design:

The system includes a clean, responsive, and user-friendly interface designed for medical professionals and patients. The UI ensures consistent navigation across dashboards for different roles and allows smooth interaction with ECG data, reports, and system notifications.

2.2.4. ECG Processing and AI Prediction Module:

The application integrates an intelligent processing pipeline that handles ECG signal/image preprocessing, feature extraction, and classification using multiple models. The system generates prediction results along with confidence scores and supports multimodal fusion for improved accuracy.

2.2.5. Backend System:

A secure and scalable backend system has been implemented using a service-based architecture. It manages ECG data storage, user authentication, prediction results, and report handling. All sensitive medical data is stored securely with proper access control and logging mechanisms.

2.2.6. Documentation:

Comprehensive documentation has been prepared, including system design, architecture diagrams, algorithms, and implementation details. This documentation supports future maintenance, upgrades, and scalability of the system while ensuring clear understanding of the project workflow.

2.3. Development Requirements

This section identifies the essential tools, technologies, and skills required for the development of the Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection. It includes programming languages, frameworks, modeling tools, and

development environments used to ensure smooth development, testing, and deployment of the system. It also ensures that the selected stack supports scalability, AI processing, and secure medical data handling. Table 2.1 shows the development tools required for the system.

Table 2.1: Development Tools Requirements

Technology	Version	Rationale
Python	3.11	Core programming language for ECG processing and ML/DL models
TensorFlow / PyTorch	2.15 / 2.2	Deep learning frameworks for model training, prediction, and multimodal fusion
OpenCV	4.9.0	ECG image preprocessing, enhancement, and noise removal
NumPy	Latest	Efficient array handling and numerical computations for signal fusion
Pandas	2.2	Data manipulation, feature structuring, and dataset organization
SciPy	1.12	Signal processing, filtering, and noise reduction in ECG data
NeuroKit2	3.2	ECG feature extraction such as peaks, intervals, and heart signal analysis
Scikit-learn	1.4	Machine learning models, classification, and feature-based prediction
Matplotlib	3.8	Data visualization and ECG signal plotting
Seaborn	Latest	Statistical visualization of classification results and model performance
SHAP / LIME	Latest	Explainable AI techniques for model interpretability
React	Latest	Frontend framework for building web-based dashboards and UI
NeuroKit2	3.2	Python library for ECG feature extraction and physiological signal analysis.
TypeScript	5.x	Statically typed programming language used to develop the React frontend with improved code reliability and maintainability.
FastAPI	0.110+	Backend framework for developing REST APIs and handling AI model inference requests.

The table 2.2 shows the Technologies with their versions and rationale required for the development.

Table 2.2: Development Technologies Requirements

Tools	Version	Rationale
VS Code	1.83	Primary IDE for frontend and backend development, debugging, and extensions
PyCharm / VS Code (Python)	Latest	Used for AI/ML model development, ECG processing, and backend scripting
StarUML	3.2.2	System modeling and UML diagram design
MS Word	Office 365	Documentation and report writing
MS PowerPoint	Office 365	Presentation preparation
Figma	116.4.2	UI/UX design and interface prototyping for dashboards
Node.js	18.x	Backend runtime environment for scalable server-side services
Postman	Latest	API testing and validation of backend services
Git & GitHub	Latest	Version control and collaborative development
Supabase / PostgreSQL	Latest	Database management, authentication, and secure data storage

2.4. Current System

Currently, various systems and platforms are used for medical data analysis, ECG interpretation, and cardiovascular disease detection. These systems differ in complexity and functionality; however, most of them share common limitations that reduce their effectiveness in real-world clinical environments. Traditional ECG analysis systems typically rely on manual interpretation or basic digital tools that only display ECG signals without advanced intelligence or decision support.

One common example is basic ECG viewer systems, which allow doctors to visualize ECG waveforms and manually interpret results. While these systems are useful for displaying patient data, they do not provide automated prediction, feature extraction, or diagnostic assistance. As a result, the accuracy and speed of diagnosis depend entirely on the expertise of the medical professional, which may lead to inconsistencies in interpretation.

Another example includes simple machine-learning based ECG classification tools, which provide automated predictions using a single model. Although these systems improve automation, they often lack multimodal integration, advanced preprocessing, and explainability features. Additionally, they do not provide proper workflow support for multiple users such as doctors, technicians, patients, and administrators, which limits their practical usability in hospital environments.

Overall, existing systems in the cardiovascular diagnostic domain are limited in terms of scalability, explainability, and workflow integration. Most solutions do not combine multiple AI models or provide a complete end-to-end clinical pipeline including ECG upload, preprocessing, prediction, and report generation. Furthermore, the absence of role-based access control and structured reporting reduces their effectiveness in real healthcare environments.

In comparison to existing systems, Table 2.3 presents an analysis of similar medical ECG-based applications and highlights their strengths and limitations. It also shows how the proposed system addresses these gaps by introducing a more advanced, integrated, and intelligent solution.

Table 2.3: Comparison of Existing System

System Name	Features	Limitations
Basic ECG Viewer System	ECG signal display, manual interpretation	No AI support, no automation, limited decision assistance
Single ML ECG Classifier [5]	Automated prediction using single model	No multimodal fusion, limited accuracy, no explainability
Hospital ECG Management System [8]	Patient record storage, ECG history tracking	No intelligent diagnosis, no predictive analytics

3. Requirement Analysis

This chapter focuses on identifying and analyzing the functional and non-functional requirements of the Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection. A clear understanding of these requirements ensures that the system is designed to be accurate, scalable, secure, and user-friendly for medical environments.

3.1. Use Cases Diagram

The Fig. 3.1 Use Case Diagram provides a comprehensive overview of the system's functionality, illustrating the interactions between different user roles and the system.

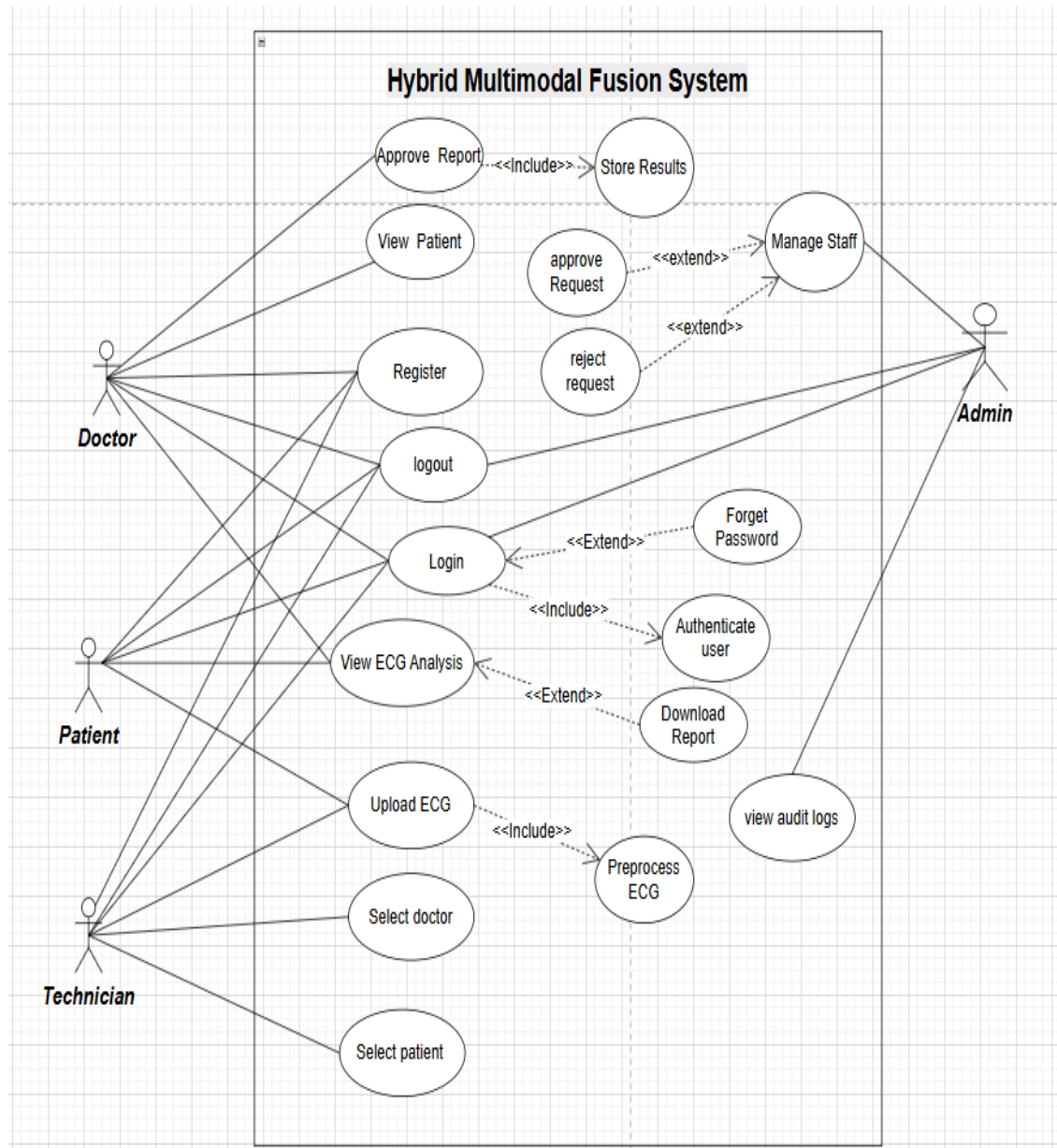


Fig. 3.1: Use Case Diagram

3.2. Detailed Use Case

A use case description provides a detailed account of how users interact with a system to achieve specific goals, outlining the scenarios, step-by-step actions, and conditions under which each interaction takes place. The following table presents the functional requirements related to user registration and authentication for the Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection system.

These requirements ensure secure access control, proper user identification, and role-based authentication for different system users including Admin, Doctor, Technician, and Patient, enabling safe and structured interaction with the platform.

3.2.1. UC-1: Register Account

The following Table 3.1 gives the detailed description of the use case Register Account.

Table 3.1 Use Case UC-01Register

Use Case ID	UC-01
Use Case Name	Register
Primary Actor	Patient (self-register), Admin (register staff), Technician (register via admin)
Secondary Actors	System, Admin (for approvals)
Description	A new user registers into the system. Patients can self-register directly, while Doctors and Technicians submit registration requests that must be approved by Admin. The system validates details, creates an account entry, triggers verification, and grants access accordingly.
Trigger	User clicks Register or Admin selects Create User in admin panel.
Preconditions	PRE-1: Registration endpoint available. PRE-2: Admin has permission to approve or reject staff accounts (if applicable).
Post Conditions	POST-1: New user account created and stored. POST-2: Verification email/SMS sent (if enabled).
Normal Flow	1. Actor opens Register page (or Admin opens Create User form). 2. Enters required details (name, email, phone, role, credentials). 3. System validates inputs (format, uniqueness). 4. System creates user record and sends verification (email/SMS) to patient. 5. For staff accounts, Admin approves (if required) and system activates account. 6. Actor receives confirmation and may log in.
Alternative Flows	A1: Social/SSO register → System links external identity and creates profile. A2: Admin approve staff and skips verification → Account activated immediately (policy dependent).
Exceptions	E1: Email/phone already in use → System shows error and suggests password reset. E2: Verification delivery fails → System allows resend.

	E3: Required fields missing/invalid → System blocks registration and highlights errors.
Business Rules	BR-1: Email must be unique per user. BR-2: Patients can self-register; staff accounts must be created/approved by Admin. BR-3: Weak passwords are rejected per password policy. BR-4: New accounts must be verified before full access is granted.
Assumptions	A1: Email/SMS service is configured and operational. A2: Admin role exists and can approve staff accounts. A3: Users have access to the email/phone they provide.

3.2.2. UC-2: Login

The following Table 3.2 gives the detailed description of the use case Login.

Table 3.2 Use Case UC-02 Login

Use Case ID	UC-02
Use Case Name	Login
Primary Actor	Doctor, Patient, Technician, Admin
Secondary Actors	System
Description	Actor enters valid credentials to access the system. The system verifies the credentials, checks role permissions, and grants access to the dashboard.
Trigger	Actor opens the application and clicks on Login.
Preconditions	PRE-1: Actor must have a registered account. PRE-2: User credentials stored in the system.
Post Conditions	POST-1: Actor logged into the system. POST-2: Actor redirected to their role-based dashboard.
Normal Flow	1. Actor opens system login page. 2. Enters username/email and password. 3. Clicks Login. 4. System verifies credentials. 5. System checks user role (Doctor/Technician/Admin/Patient). 6. Actor successfully logs in and is redirected to dashboard.
Alternative Flows	A1: Actor clicks Forgot Password → System sends reset link. A2: Actor enters wrong credentials → System displays error and retries allowed.
Exceptions	E1: Account locked due to multiple failed attempts. E2: Server connection error during login. E3: User role not assigned → Access denied.
Business Rules	BR-1: Password must meet security standards (min length, complexity). BR-2: Only authenticated users can access any internal module.
Assumptions	A1: Authentication server is working.

	A2: User has stable internet connection.
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3.2.3. UC-3: Logout User

The following Table 3.3 gives the detailed description of the use case Logout User.

Table 3.3 Use Case UC-03 Logout User

Use Case ID	UC-03
Use Case Name	Logout
Primary Actor	Any authenticated user Doctor ,Patient , Technician , Admin
Secondary Actor	Authentication Server / Session Manager
Description	User terminates the current logged-in session. The system securely invalidates the active authentication token, clears session variables, removes temporary access rights, and returns the user to the Login page. This ensures security, privacy, and prevents unauthorized reuse of session credentials.
Trigger	User taps/clicks Logout from navigation menu or profile dropdown.
Preconditions	• User must be successfully logged in. • Session token or JWT must be active. • System internet connectivity is stable.
Postconditions	• Session is terminated immediately. • All temporary buffers/sessions cleared. • User lands on Login screen and must reauthenticate to re-enter.
Normal Flow	1. User selects Logout. 2. System prompts a confirmation (optional). 3. Upon user confirmation, backend invalidates JWT/session token. 4. Access tokens, cookies, and cached authentication values are deleted. 5. User receives logout success notification. 6. System redirects user to Login page.
Alternative Flow	AF-01 Cancel Logout: • User cancels confirmation dialogue → session continues normally. AF-02 Auto Logout: • Session expires automatically after inactivity timeout → system redirects to Login.
Exceptions	E-01 Session expired before manual logout → redirect occurs without confirmation.
Business Rules	• A user cannot access any authorized page after logout. • Session timeout policy applies based on role/security level.
Assumptions	• User intends to end session willingly. • Logout does not erase stored profile, ECG records or medical history.

3.2.4. UC-4: View ECG Analysis

The following Table 3.4 gives the detailed description of the use case View ECG Analysis.

Table 3.4 Use Case UC-04 View ECG Analysis

Use Case ID	UC-04
Use Case Name	View ECG Analysis
Primary Actor	Doctor
Secondary Actors	Patient
Description	Actor opens the detailed ECG analysis page showing raw ECG waveforms, spectrograms, feature summaries, model predictions, confidence scores, and explainability overlays. This helps clinicians interpret AI findings alongside raw signals.
Trigger	Actor selects an ECG record and clicks View Analysis.
Preconditions	PRE-1: Actor is authenticated and authorized. PRE-2: ECG record and analysis results are available and linked to the patient.
Post Conditions	POST-1: ECG analysis page is displayed with all relevant visualizations and metadata. POST-2: System logs the access for auditing.
Normal Flow	1. Actor logs in and navigates to patient's ECG list. 2. Selects the desired ECG record. 3. Clicks View Analysis. 4. System loads raw waveform, spectrogram image, extracted features, AI prediction(s) with confidence, and XAI highlights 5. Actor reviews results
Alternative Flows	A1: Actor toggles between leads or time windows A2: Actor requests raw data download
Exceptions	E1: Analysis results not yet available → System shows "Processing" and option to notify when ready. E2: Visualization components fail to render (browser/format issue) → System provides fallback text summary. E3: Unauthorized access → System denies access and logs the attempt.
Business Rules	BR-1: Only authorized personnel can view full
Assumptions	A1: AI engine has completed analysis and saved outputs. A2: Visualization libraries are supported in the user's browser.

3.2.5. UC-5: Upload ECG

The following Table 3.5 gives the detailed description of the use case Upload ECG.

Table 3.5 Use Case UC-05 Upload ECG

Use Case ID	UC-05
Use Case Name	Upload ECG
Primary Actor	Technician, Patient
Secondary Actors	System (AI Engine), Doctor
Description	Technician or Patient uploads an ECG file (image or signal). The system validates the file, stores it, and sends it for AI processing. Doctor is notified once the analysis is completed.
Trigger	Actor clicks Upload ECG and selects a file.
Preconditions	PRE-1: Actor (Technician/Patient) is logged in. PRE-2: Patient record must exist (or patient is self-uploading). PRE-3: ECG file must be in supported format (PNG/JPG/ etc.).
Post Conditions	POST-1: ECG successfully uploaded and stored. POST-2: AI engine receives ECG for processing. POST-3: Doctor notified when analysis is completed.
Normal Flow	1. Actor logs in. 2. Opens Upload ECG page. 3. Selects patient (Technician) OR system auto-selects (Patient). 4. Uploads ECG file. 5. System validates format and quality. 6. System sends ECG to AI analysis queue. 7. Actor receives confirmation; doctor gets notification once processed.
Alternative Flows	A1: Technician selects a different patient before upload.
Exceptions	E1: Unsupported file format → System rejects and shows message. E2: File corrupted → System blocks upload. E3: Patient not selected (Technician only) → Upload blocked.
Business Rules	BR-1: Patients can upload only their own ECGs; Technicians can upload for any assigned patient. BR-2: All uploads must be linked with a verified patient ID. BR-3: Maximum file size and approved formats enforced.
Assumptions	A1: AI engine available and running. A2: Storage server has enough capacity. A3: Patient uploads are clinically valid (from verified ECG machine or lab).

3.2.6. UC-6: Approve Report

The following Table 3.6 gives the detailed description of the use case Approve Report

Table 3.6 Use Case UC-06 Approve Report

Use Case ID	UC-06
Use Case Name	Approve Report
Primary Actor	Doctor
Secondary Actors	Technician, System, Patient
Description	Doctor reviews an ECG report generated by the system. After reviewing, the doctor approves the report. The system stores the approved report and notifies relevant users.
Trigger	Doctor selects a pending ECG report for review.
Preconditions	PRE-1: Doctor is logged in. PRE-2: ECG report exists in “Pending” status. PRE-3: Patient record exists.
Post Conditions	POST-1: Report status updated to “Approved”. POST-2: Approved report stored in system. POST-3: Notifications sent to Technician/Patient.
Normal Flow	1. Doctor logs in. 2. Opens list of pending reports. 3. Selects and views a report. 4. Reviews analysis and explanations. 5. Clicks Approve. 6. System stores the approved report and sends notifications.
Alternative Flows	A1: Doctor requests changes → Status becomes “Needs Revision”. A2: Doctor rejects report → Status becomes “Rejected”.
Exceptions	E1: Report file unreadable. E2: Patient record missing or unlinked. E3: Error saving approval due to server/storage failure.
Business Rules	BR-1: Only authorized doctors may approve reports. BR-2: Every approval must include timestamp & doctor ID.
Assumptions	A1: AI-generated report already available. A2: System storage and network working normally.

3.2.7. UC-7: View Patient

The following Table 3.7 gives the detailed description of the use case View Patient.

Table 3.7 Use Case UC-07 View Patient

Use Case ID	UC-07
Use Case Name	View Patient
Primary Actor	Doctor, Technician

Secondary Actors	System
Description	The actor opens and views a patient's complete profile, including demographics, medical history, uploaded ECG records, and previously approved reports. This helps in diagnosis and decision-making.
Trigger	Actor selects a patient from the patient list or search results.
Preconditions	PRE-1: Actor is logged in. PRE-2: Patient record exists. PRE-3: Actor is authorized to access this patient's data.
Post Conditions	POST-1: Patient details displayed correctly. POST-2: System logs access for audit purposes.
Normal Flow	1. Actor logs in. 2. Navigates to Patient Management 3. Searches or selects a patient. 4. System retrieves and displays patient profile. 5. Actor reviews ECG history, reports, and details.
Alternative Flows	A1: Patient has no ECG history → System shows "No Records Found". A2: Actor filters patients (by ID, name, date).
Exceptions	E1: Patient record not found. E2: Actor not authorized to view this patient. E3: Server/database error while fetching data.
Business Rules	BR-1: Doctors may view only their assigned patients. BR-2: Every patient view event must be logged.
Assumptions	A1: Patient data is stored and updated correctly. A2: System search and filtering functions are working.

3.2.8. UC-8: Download Report

The following Table 3.8 gives the detailed description of the use case Download Report.

Table 3.8 Use Case UC-08 Download Report

Use Case ID	UC-08
Use Case Name	Download Report
Primary Actor	Doctor, Patient
Secondary Actors	System
Description	Actor downloads an approved ECG report in PDF or image format for review, printing, or record-keeping. System retrieves the stored report and provides it as a downloadable file.
Trigger	Actor clicks the "Download Report" button on a patient's report page.
Preconditions	PRE-1: Actor is logged in (Doctor/Patient). PRE-2: Report exists and is in Approved state.

	PRE-3: Actor has permission to access the report.
Post Conditions	POST-1: Report file downloaded successfully. POST-2: Download event logged in system audit.
Normal Flow	1. Actor logs in. 2. Navigates to patient report section. 3. Selects a specific approved report. 4. Clicks Download Report. 5. System retrieves stored PDF/image. 6. File is downloaded to the actor's device.
Alternative Flows	A1: Actor chooses different format (PDF/Image) → System generates selected format.
Exceptions	E1: Report not approved → System shows "Report not available for download". E2: File missing or corrupted on server. E3: No permission → "Access Denied" message.
Business Rules	BR-2: All downloads must be recorded with time and user ID. BR-3: Doctor can download all assigned patient reports; patient can download only their own reports.
Assumptions	A1: Report files are stored correctly. A2: System supports secure file download.

3.2.9. UC-9: View Audit Logs

The following Table 3.9 gives the detailed description of the use case View Audit Logs.

Table 3.9 Use Case UC-09 View Audit Logs

Use Case ID	UC-10
Use Case Name	View Audit Logs
Primary Actor	Admin
Secondary Actor	System Database / Log Manager
Description	Admin accesses and reviews stored audit logs related to system activities such as login/logout events, report uploads, approvals, patient data access, and configuration changes. The audit module retrieves records chronologically, supports filters (date, user, action type), and displays results for monitoring and forensic traceability.
Trigger	Admin selects "Audit Logs" option from the Admin Dashboard.
Preconditions	• Admin must be authenticated. • Audit logging service must be running. • Logs must exist in storage/database.
Postconditions	• Audit logs are successfully displayed to admin. • Optional: Admin may export or filter logs for further review.
Normal Flow	1. Admin navigates to Audit Logs page. 2. System fetches logs from log storage or database.

	3. Logs are displayed in tabular/visual format. 4. Admin applies filters (date/user/module/type) if needed. 5. System refines and presents filtered results. 6. Admin reviews and monitors suspicious or important events.
Alternative Flow	AF-01 No Logs Available: • System displays "No Audit Data Found".
Exceptions	E-01: Database Connection Failure: system shows error and provides retry option. E-02: Log File Corruption: missing or unreadable entries notified to admin.
Business Rules	• Only Admin role has permission to view audit logs. • Logs must be stored securely and tamper-proof.
Assumptions	• Logs are updated in real-time. • Admin regularly monitors logs for security, misuse, and compliance.

3.2.10. UC-10: Manage Staff

The following Table 3.10 gives the detailed description of the use case Manage Staff.

Table 3.10 Use Case UC-10 Manage Staff

Use Case ID	UC-10
Use Case Name	Manage Staff
Primary Actor	Admin
Secondary Actors	System, Technician, Doctor
Description	Admin manages staff accounts and roles: create, update, deactivate, assign roles/permissions, and view staff details. Ensures correct access control and personnel records for system operations.
Trigger	Admin selects Manage Staff from the admin dashboard.
Preconditions	PRE-1: Admin is authenticated and authorized. PRE-2: Staff management module accessible.
Post Conditions	POST-1: Staff records updated/created/deactivated as requested. POST-2: Role and permission changes take effect. POST-3: Audit log entry created for the action.
Normal Flow	1. Admin logs in and opens Manage Staff. 2. Admin views staff list or searches for a staff member. 3. Admin selects action: Add New, Edit, Assign Role, Deactivate, or View. 4. Admin fills form or selects roles/permissions. 5. System validates inputs and saves changes.
Alternative Flows	A1: Admin delegate's role assignment to HR → System records delegate and approval.
Exceptions	E1: Duplicate staff identifier (email/ID) → System rejects creation.
Business Rules	BR-1: Only Admin role may create or delete staff accounts. BR-2: Role changes require approval if they elevate privileges. BR-3: Staff must have unique email/ID.

Assumptions	A1: Admin has authority for staffing actions.
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3.2.11. UC-11: View Report

The following Table 3.11 gives the detailed description of the use case View Report.

Table 3.11 Use Case UC-11 View Report

Use Case ID	UC-11
Use Case Name	View Report
Primary Actor	Doctor, Patient
Secondary Actors	Admin
Description	Actor opens and views a finalized ECG report (approved, rejected, or pending) including summary, detailed findings, attached files AI annotations, and clinician notes. The view allows printing, commenting, or initiating next actions (download, approve, request changes).
Trigger	Actor selects a report from patient history, notifications, or search results and clicks View Report.
Preconditions	PRE-1: Actor is authenticated and authorized. PRE-2: Target report exists and is linked to the patient. PRE-3: Storage service is available to fetch report files.
Post Conditions	POST-1: Report is displayed with metadata, analysis, and audit log entry created for access. POST-2: Actor may take follow-up actions (download, comment, and approve/request changes) which are routed to appropriate workflows.
Normal Flow	1. Actor logs in and navigates to patient reports. 2. Selects desired report (by date, ID, or notification). 3. System retrieves report summary, full content (PDF/image), AI analysis results, and clinician notes. 4. System displays the report with explainability overlays and metadata (creator, status, timestamps). 5. Actor reviews content and may choose actions: Download, Comment, and Approve/Request Changes (if authorized).
Alternative Flows	A1: Actor toggles between compact/expanded view → System adjusts displayed detail level. A2: Actor filters report content (show only AI highlights or clinician notes). A3: Actor opens an older version or amendment → System shows version history and differences.
Exceptions	E1: Report file missing or corrupted → System shows error and offers fallback summary or requests reupload. E2: Actor not authorized for this report → Access denied and attempt logged. E3: Network/storage timeout → System shows “Unable to load” with retry option.

Business Rules	BR-1: Only authorized users may view reports; patients can view only their own. BR-2: Every view must be recorded for audit with user ID and timestamp. BR-4: Approved reports are read-only; comments or amendments follow defined workflows.
Assumptions	A1: Report metadata is correct and linked to the right patient. A2: Visualization and PDF rendering supported in user's browser/device.

3.2.12. UC-12: Select Patient

The following Table 3.12 gives the detailed description of the use case Select Patient.

Table 3.12 Use Case UC-12 select patient

Use Case ID	UC-12
Use Case Name	Select Patient
Primary Actor	Technician
Secondary Actor	System / Patient Database
Description	The actor searches and selects a specific patient from the system in order to upload ECG data, view existing records, or link a report to that patient. The system retrieves patient profiles based on search criteria (ID, name, CNIC, MRN, etc.) and displays a list from which the actor chooses one patient to continue the workflow.
Trigger	Technician/Doctor clicks "Select Patient" from the relevant screen (e.g., Upload ECG, View Patient, Approve Report).
Preconditions	• Actor must be logged into the system. • Patient records must already exist in the database. • Actor must have permission to access patient data.
Postconditions	• A specific patient is selected and marked as "current context patient". • Subsequent actions (Upload ECG, View ECG Analysis, Approve Report, etc.) are linked to this selected patient.
Normal Flow	1. Actor opens the Select Patient screen. 2. System displays a search box and/or a list of patients. 3. Actor enters search criteria (e.g., name, patient ID, contact, CNIC). 4. System queries the patient database and returns matching results. 5. System displays a list/table of matched patients. 6. Actor reviews the list and clicks on the appropriate patient. 7. System confirms the selection and sets this patient as the active patient for the ongoing task.
Alternative Flow	AF-01 No Match Found: • 4a. No patient matches the search criteria. • 4b. System displays "No patient found" message and suggests refining the search.

Exceptions	E-01: Database connection error while fetching patient's → system shows error message and asks user to retry. E-02: Actor tries to select a patient without required permissions → system denies access and logs the incident.
Business Rules	• Only authorized roles (Technician/Doctor/Admin as per policy) may view and select patients. • Patient data shown must comply with privacy and security regulations (e.g., minimal necessary info). • Every selection action may be logged in audit logs for traceability.
Assumptions	• Patient data in the system is accurate and up to date. • The selected patient will be used immediately in a subsequent use case (e.g., Upload ECG, View/Approve Report).

3.2.13. UC-13: Select Doctor

The following Table 3.13 gives the detailed description of the use case Select Doctor.

Table 3.13 Use Case UC-13 select doctor

Use Case ID	UC-08
Use Case Name	Select Doctor
Primary Actor	Technician
Secondary Actor	Doctor Database / System Backend
Description	The user selects a doctor from the available list in order to assign ECG reports, request diagnosis, or view medical feedback. The system retrieves doctor profiles, displays names, specialties, availability, and allows the user to choose one doctor to proceed with further clinical workflow.
Trigger	Patient/Technician clicks "Select Doctor" from Upload ECG, Book Appointment, or Report Review interface.
Preconditions	• Actor must be logged in. • Doctor profiles must exist in the system. • Actor must have the permission to assign/select a doctor.
Postconditions	• A doctor is successfully selected and linked with the patient/test/report. • The selected doctor becomes the active doctor context for further workflow like ECG Upload, Report Review, or Approval handling.
Normal Flow	1. Actor opens the Select Doctor page. 2. System displays all registered doctors in a list/grid. 3. Actor searches doctor by name, specialty, experience, or availability. 4. System fetches matching doctors and displays results. 5. Actor selects one doctor from the result list. 6. System confirms selection and stores the link for upcoming actions.
Alternative Flow	AF-01 No Doctor Available • System displays "No doctors found" and suggests retry/another search.

Exceptions	E-01 Database Failure — Doctors cannot be fetched → system displays error & retry option..
Business Rules	• Only verified doctors appear for selection.
Assumptions	• Doctor data (specialty, availability, profile info) is up-to-date

3.3. Functional Requirements

This section lists the core functions that the Hybrid Fusion Framework must perform, detailing the specific features and behaviours required to fulfil the needs of users and stakeholder.

3.3.1. User Registration:

As shown in Table 3.14, FR-1 User Registration is mapped to its corresponding design component.

Table 3.14 FR-1 User Registration

Identifier	FR-1
Title	User Registration
Requirement	The system shall allow new users to register by entering their name, email, phone number, role, and password. The system must validate format, uniqueness, and store the account securely. Staff accounts require admin approval while patients can self-register.
Source	UC-01
Rationale	Required to create accounts and enable user access to system services.
Business Rule	BR-1: Email must be unique. BR-2: Staff must be admin-approved. BR-3: Weak passwords rejected. BR-4: Account verification required.
Dependencies	FR-2 (Login)
Priority	High

3.3.2. User Login:

As shown in Table 3.15, FR-2 User Login is mapped to its corresponding design component.

Table 3.15 FR-2 User Login

Identifier	FR-2
Title	User Login
Requirement	The system shall authenticate users based on email/username and password. On successful authentication, system redirects the user to a role-based dashboard.
Source	UC-02
Rationale	Provides secure access to system features.
Business Rule	BR-1: Password must meet security guidelines. BR-2: Only authenticated users may access internal modules.
Dependencies	FR-1 (Registration)

3.3.3. Logout:

As shown in Table 3.16, FR-3 Logout is mapped to its corresponding design component.

Table 3.16 FR-3 Logout

Identifier	FR-3
Title	Logout
Requirement	The system shall allow users to securely terminate their session. Authentication tokens, cookies, and session variables must be invalidated.
Source	UC-03
Rationale	Prevents unauthorized access through expired or open sessions.
Business Rule	Session timeout policies apply according to security level.
Dependencies	FR-2 (Login)

3.3.4. View ECG Analysis:

As shown in Table 3.17, FR-4 View ECG Analysis is mapped to its corresponding design component.

Table 3.17 FR-4 View ECG Analysis

Identifier	FR-4
Title	View ECG Analysis
Requirement	The system shall allow doctors to view raw ECG waveform, spectrograms, AI-generated prediction, confidence score, and explainability overlays.
Source	UC-04
Rationale	Supports clinical diagnosis and decision-making.
Business Rule	Access allowed only to authorized doctors.
Dependencies	FR-5 (Upload ECG), FR-6 (Approve Report)
Priority	High

3.3.5. Upload ECG:

As shown in Table 3.18, FR-5 Upload ECG is mapped to its corresponding design component.

Table 3.18 FR-5 Upload ECG

Identifier	FR-5
Title	Upload ECG
Requirement	Patients or Technicians shall upload ECG files in supported formats. The system validates file type, size, links patient ID, stores ECG and forwards for AI processing.
Source	UC-05
Rationale	Essential for diagnosis workflow and system analysis.

Business Rule	Only PNG/JPG supported. Must be linked to patient. Size limit enforced.
Dependencies	FR-1 (User Profile), FR-4 (ECG Analysis)
Priority	High

3.3.6. Approve ECG Report:

As shown in Table 3.19, FR-6 Approve ECG Report is mapped to its corresponding design component.

Table 3.19 FR-6 Approve ECG Report

Identifier	FR-6
Title	Approve ECG Report
Requirement	The system shall allow doctors to approve, request changes, or reject ECG reports. Approval must trigger notifications.
Source	UC-06
Rationale	Enables clinical validation of AI report before use.
Business Rule	Only authorized doctors may approve. Timestamp + Doctor ID stored.
Dependencies	FR-4 (View Analysis), FR-5 (Upload ECG)
Priority	High

3.3.7. View Patient details:

As shown in Table 3.20, FR-7 View Patient Details is mapped to its corresponding design component.

Table 3.20 FR-7 View Patient Details

Identifier	FR-7
Title	View Patient Details
Requirement	The system shall allow authorized users to view patient demographics, ECG history, reports, and medical profile.
Source	UC-07
Rationale	Allows diagnosis and clinical review.
Business Rule	Doctors view assigned patients only. Access logged.
Dependencies	FR-1, FR-11

3.3.8. Download ECG Report:

As shown in Table 3.21, FR-8 Download ECG Report is mapped to its corresponding design component

Table 3.21 FR-8 Download ECG Report

Identifier	FR-8
Title	Download ECG Report
Requirement	The system shall allow patients/doctors to download approved ECG reports in PDF/Image format.
Source	UC-08

Rationale	Needed for printing, storage and external sharing.
Business Rule	Only approved reports downloadable. Download event logged.
Dependencies	FR-6 (Approval)

3.3.9. View Audit Logs:

As shown in Table 3.22, FR-9 View Audit Logs is mapped to its corresponding design component.

Table 3.22 FR-9 View Audit Logs

Identifier	FR-9
Title	View Audit Logs
Requirement	The system shall allow admins to view filtered audit logs by date, user or event type for monitoring activities.
Source	UC-09
Rationale	Ensures system security and forensic traceability.
Business Rule	Only Admins allowed. Logs tamper-proof.
Dependencies	FR-2 (Login)
Priority	High

3.3.10. Manage Staff Accounts:

As shown in Table 3.23, FR-10 Manage Staff Accounts is mapped to its corresponding design component.

Table 3.23 FR-10 Manage Staff Accounts

Identifier	FR-10
Title	Manage Staff Accounts
Requirement	The system shall allow admin to add/edit/remove staff, assign roles, and modify access permissions.
Source	UC-10
Rationale	Required for controlled authorization and personnel management.
Business Rule	Staff ID/email must be unique. Role escalation requires approval.
Dependencies	FR-1 (Registration)
Priority	High

3.3.11. View ECG Report:

As shown in Table 3.24, FR-11 View ECG Report is mapped to its corresponding design component.

Table 3.24 FR-11 View ECG Report

Identifier	FR-11
Title	View ECG Report
Requirement	The system shall allow authorized users to view full report with AI highlights, notes, metadata & status.
Source	UC-11

Rationale	Required for patient review & clinical finalization.
Business Rule	Approved reports read-only. View events logged.
Dependencies	FR-6 (Approve Report)
Priority	High

3.3.12. Select Patient:

As shown in Table 3.25, FR-12 Select Patient is mapped to its corresponding design component.

Table 3.25 FR-12 Select Patient

Identifier	FR-12
Title	Select Patient
Requirement	The system shall allow technicians/doctors to search and select a patient using name, ID, CNIC etc.
Source	UC-12
Rationale	Needed before performing ECG upload or report actions.
Business Rule	Only authorized users can view/select.
Dependencies	FR-7 (View Patient)
Priority	High

3.3.13. Select Doctor:

As shown in Table 3.26, FR-13 Select Doctor is mapped to its corresponding design component

Table 3.26 FR-13 Select Doctor

Identifier	FR-13
Title	Select Doctor
Requirement	The system shall allow selecting a doctor based on name, specialty or availability for report assignment or consultation.
Source	UC-13
Rationale	Allows routing ECG reports for diagnosis.
Business Rule	Only verified doctors appear for assignment.
Dependencies	FR-10 (Manage Staff)
Priority	Medium/High

3.4. Non-Functional Requirements

This section specifies the quality attributes of the Hybrid Fusion Framework CVD including usability, performance, security, reliability, and other criteria that ensure a robust and user-friendly system.

3.4.1. Usability

Non-functional requirements for usability focus on how easy a system is to use,

including ease of learning, intuitive navigation, and overall user experience.

- **USE-1:** The system shall allow a technician to upload an ECG within 3 clicks for quick workflow.
- **USE-2:** A doctor shall be able to view new ECG results from the dashboard in a single click for faster review.
- **USE-3:** A patient shall access their verified report in 2 steps to ensure ease of use.
- **USE-4:** Clear and simple error messages shall appear for failed or invalid uploads to guide users.
- **USE-5:** UI text shall maintain a minimum size of 14px to ensure readability for all users.

3.4.2. Performance

Performance specifies how quickly a system responds to requests, how efficiently it uses resources, and how it handles different workloads

- **PER-1:** Web pages shall load in under 3 seconds on a normal 10 Mbps connection for smooth experience.
- **PER-2:** ECG preprocessing shall complete within 5 seconds to avoid processing delays.
- **PER-3:** AI prediction with explain ability shall take around 10 seconds to provide quick results.
- **PER-4:** PDF report generation shall complete within 5 seconds for rapid access and download.
- **PER-5:** The system shall support up to 50 active users simultaneously without slowdown.

3.4.3. Reliability

Reliability ensures the system operates consistently without failure, maintaining accurate and stable performance. Availability ensures the system remains accessible and functional whenever required.

- **REL-1:** The system shall maintain at least 95% uptime during normal operating hours for availability.
- **REL-2:** ECG processing shall retry automatically twice if initial attempt fails to ensure completion.
- **REL-3:** All important actions shall be logged with timestamp and user ID for traceability.

3.4.4. Security

Security ensures that user data and system resources are protected against unauthorized access, breaches, and misuse.

- **SEC-1:** The system shall enforce role-based access so users only access permitted sections.
- **SEC-2:** All data transmission shall occur over HTTPS for secure communication.

3.4.5. Maintainability

Maintainability ensures the system can be easily updated, fixed, and improved with minimal effort.

- **MAINT-1:** The system shall use modular structure to update or replace components easily.
- **MAINT-2:** Code shall follow standard development conventions to keep the system understandable.

3.4.6. Scalability

Scalability ensures the system can handle increasing users, data, and workload without degrading performance.

- **SCA-1:** The AI processing system shall allow adding multiple workers for higher load handling.

3.4.7. Portability

Portability ensures the system can run across different platforms and environments without major changes.

- **PORT-1:** The system shall operate on Windows.
- **PORT-2:** The system shall support major browsers including Chrome, Firefox, Edge, and Safari.

3.4.8. Data Integrity

Data Integrity ensures that all data within the system remains accurate, consistent, and reliable throughout its lifecycle, preventing unauthorized modification, loss, or corruption.

- **DATA-1:** Each ECG record shall remain permanently linked to the correct patient profile.
- **DATA-2:** Model version, time of prediction, and confidence score shall be saved for each result.

4. Design and Architecture

This chapter describes the architecture and design of the Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection. It includes system architecture, process flow, and various design models that define the structure and functionality of the system, ensuring smooth interaction between the client, server, and AI services for efficient ECG-based diagnosis and reporting.

4.1. Architecture Design

The proposed system follows a Micro services architecture for our Hybrid Multimodal Fusion System designed to analyze ECG data and generate AI-assisted cardiovascular disease predictions. The API Gateway acts as a central entry point, forwarding all client requests to their respective services. The Authentication Service manages secure login and role-based access, while the User & Role Management Service maintains user accounts and permissions. The Patient Management Service handles patient records and doctor–patient assignments. The ECG Upload Service receives ECG files, validates them, and stores metadata for processing. The Preprocessing Service prepares ECG data through denoising and segmentation, followed by the Inference Service, which performs machine-learning and deep-learning based predictions. The XAI Service generates explainability visuals to support clinical decision-making, and the Report Generation Service produces doctor-verified medical reports. The Fig. 4.1 shows that all services store and retrieve structured data from the database, ensuring reliable, scalable, and efficient operation of the ECG diagnostic workflow.

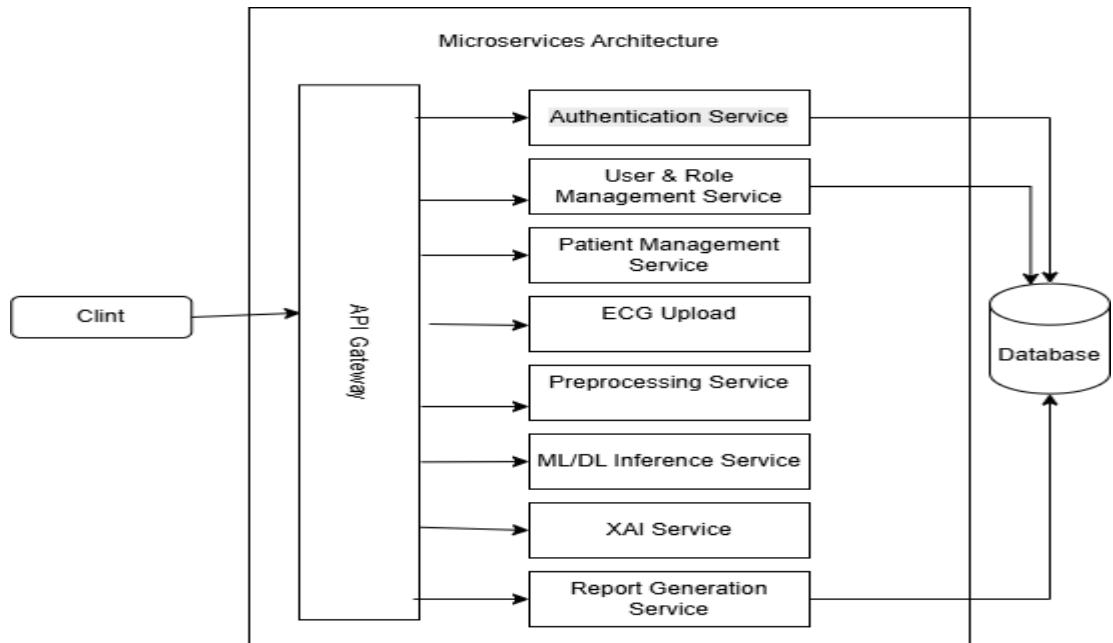


Fig. 4.1: Micro Services Architecture

4.2. Process Flow/Representation

This section presents a high-level representation of the system workflow, showing how all actors including doctors, technicians, patients, and admins interact throughout the diagnostic process. It outlines the flow of operations such as ECG upload, preprocessing, AI-based inference, XAI visualization, and final report generation. Overall, it provides a concise view of how data and services connect to complete the cardiovascular diagnosis cycle.

4.2.1. Doctor Process Flow

Fig. 4.2 presents the doctor's flow from login, viewing patients or pending reports, evaluating ECG results, and approving or rejecting them, followed by system notifications.

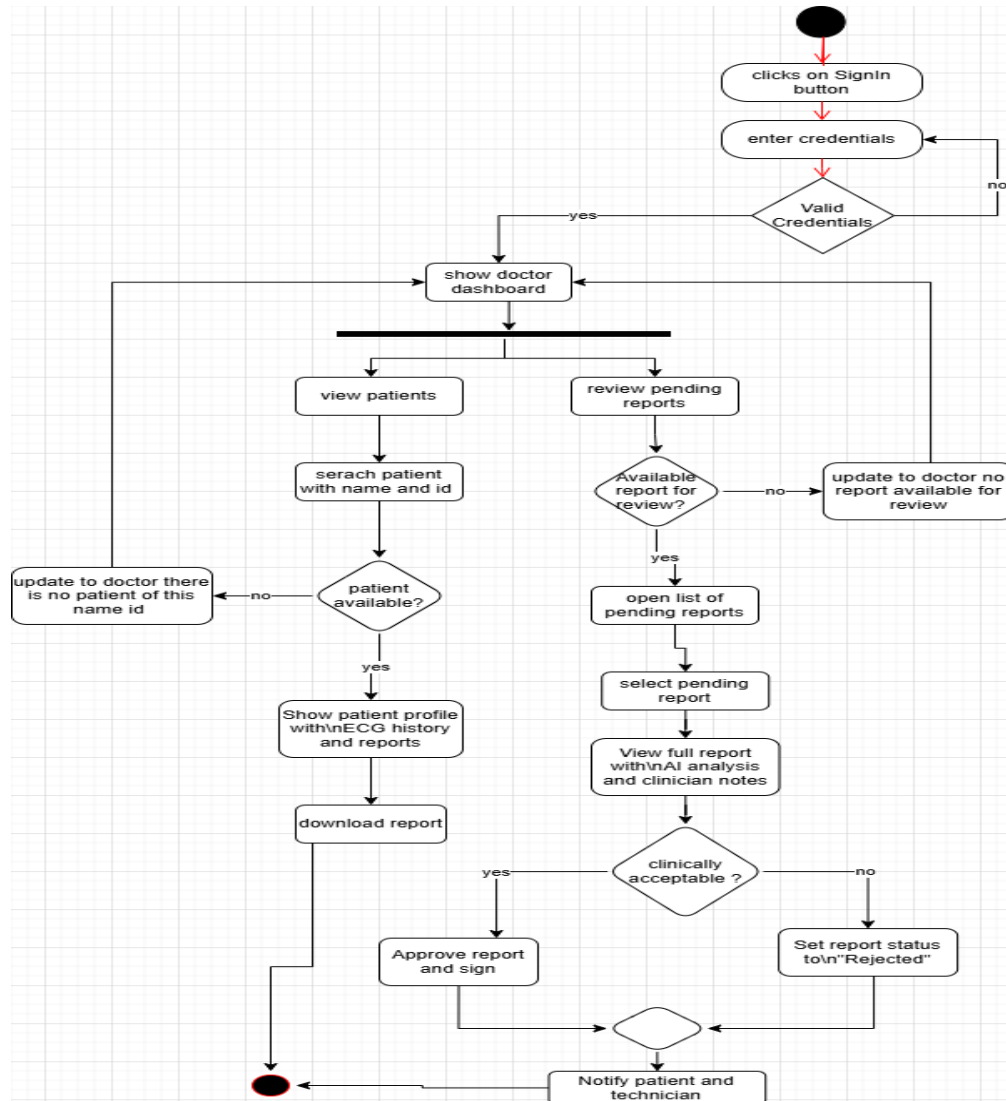


Fig. 4.2: Process Flow Representation for Doctor

4.2.2. Technician Process Flow

Fig. 4.3 shows the technician workflow from login and dashboard access to viewing patient history or selecting a patient and doctor for ECG upload. After choosing a valid file, the system verifies format and size, uploads it successfully, and notifies the assigned doctor for review.

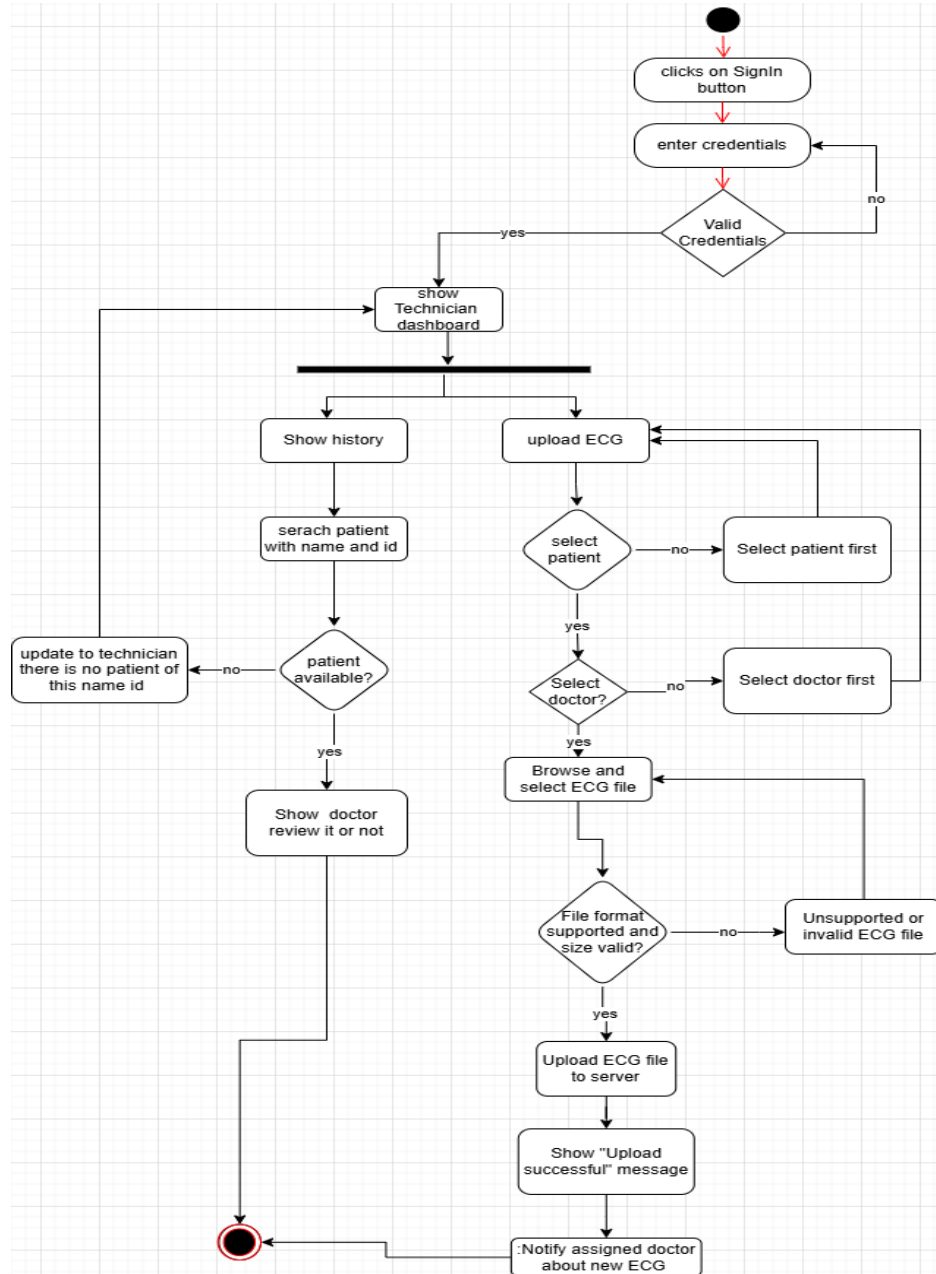


Fig. 4.3: Process Flow Representation for Technician

4.2.3. Patient Process Flow

Fig. 4.4 shows the patient flow from login to viewing reports or uploading a new ECG. Valid files are processed and sent for AI diagnosis, while reports can later be viewed or downloaded. Invalid uploads or unavailable reports trigger notifications.

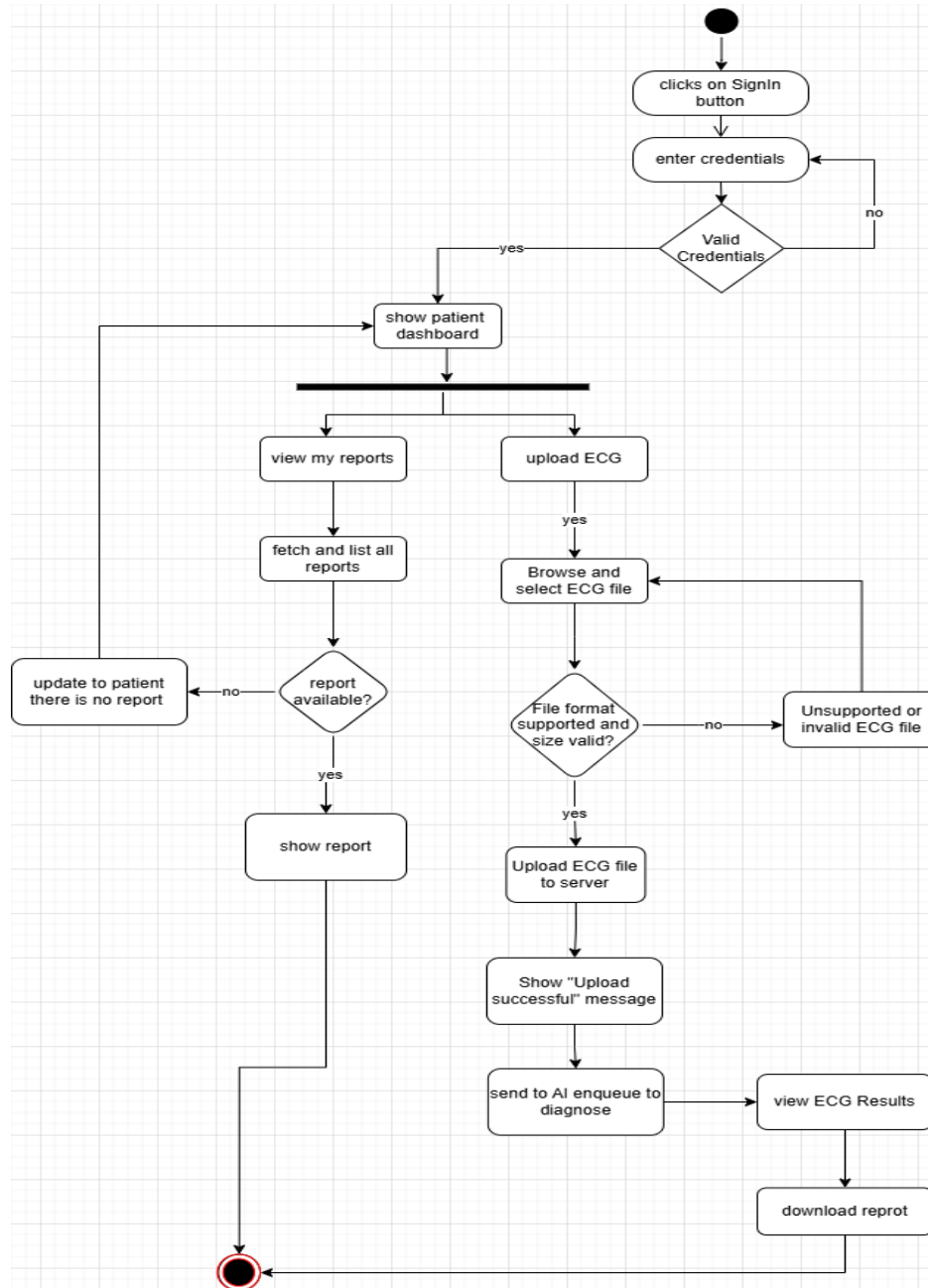


Fig. 4.4: Process Flow Representation for Patient

4.2.4. Admin Process Flow

Fig. 4.5 represents the admin workflow from login to dashboard access, where the admin can view audit logs or handle user approval requests. Audit logs can be filtered and displayed, while pending user accounts may be approved or rejected, enabling or denying system access accordingly.

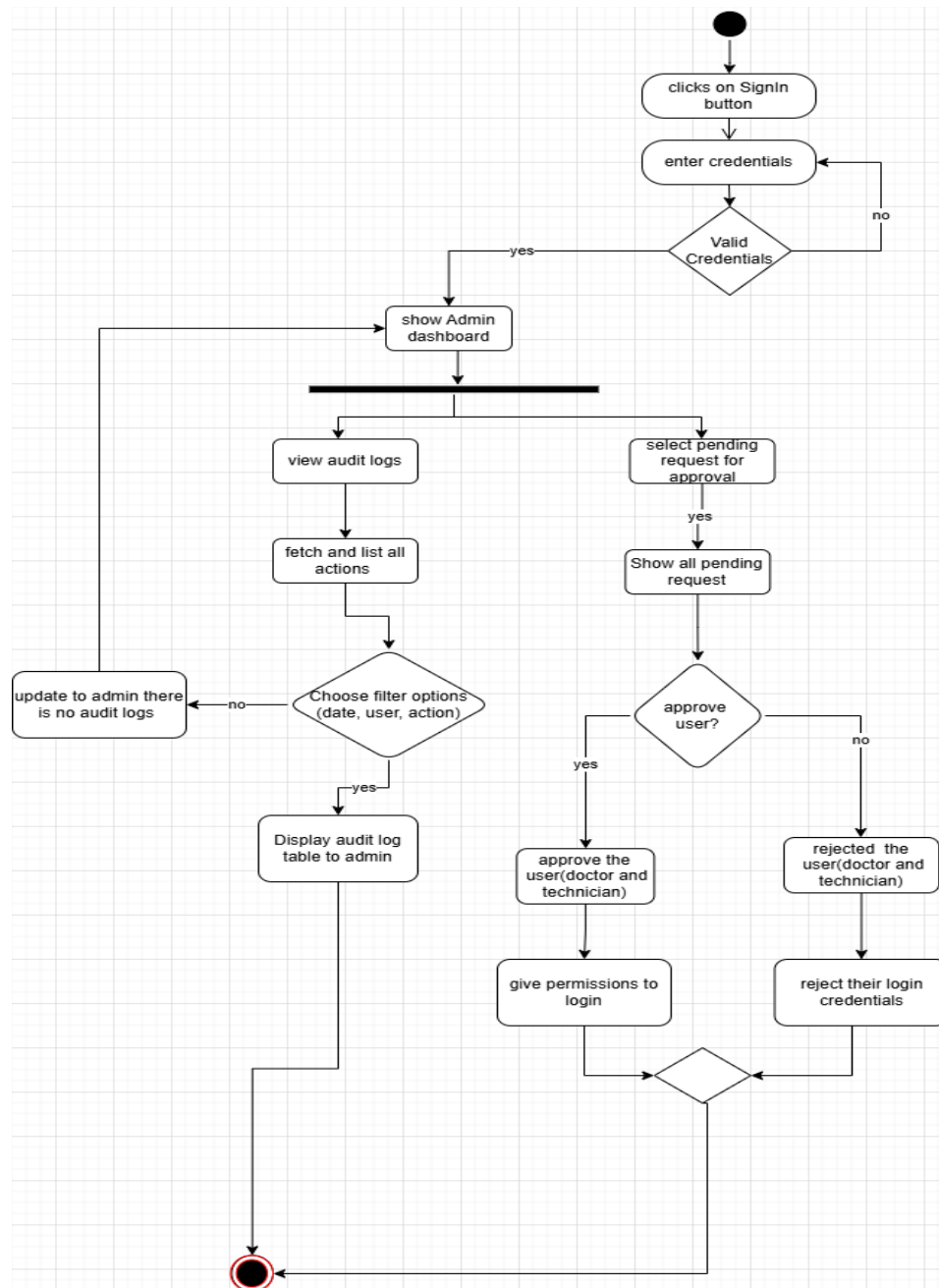


Fig. 4.5: Process Flow Representation for Admin

4.3. Design Models

This section presents the visual and structural models used to describe system behavior and architecture. These models help illustrate interactions, decision flows, and the internal class structure of the Hybrid Multimodal Fusion System

4.3.1. System Sequence diagram

The System Sequence Diagram illustrates the ordered exchange of messages between the primary actors Doctor, Patient, Technician and the Hybrid Multimodal Fusion System. It demonstrates how user requests are processed sequentially, including authentication, ECG uploading, report retrieval and clinical decision approval,. The diagram also reflects the system's internal communication with AI services and the database, ensuring smooth execution of diagnostic and management workflows.

4.3.1.1 Doctor System Sequence Diagram

Fig. 4.6 illustrates the doctor system interaction flow, covering authentication, patient selection, ECG diagnosis, and report approval or rejection.

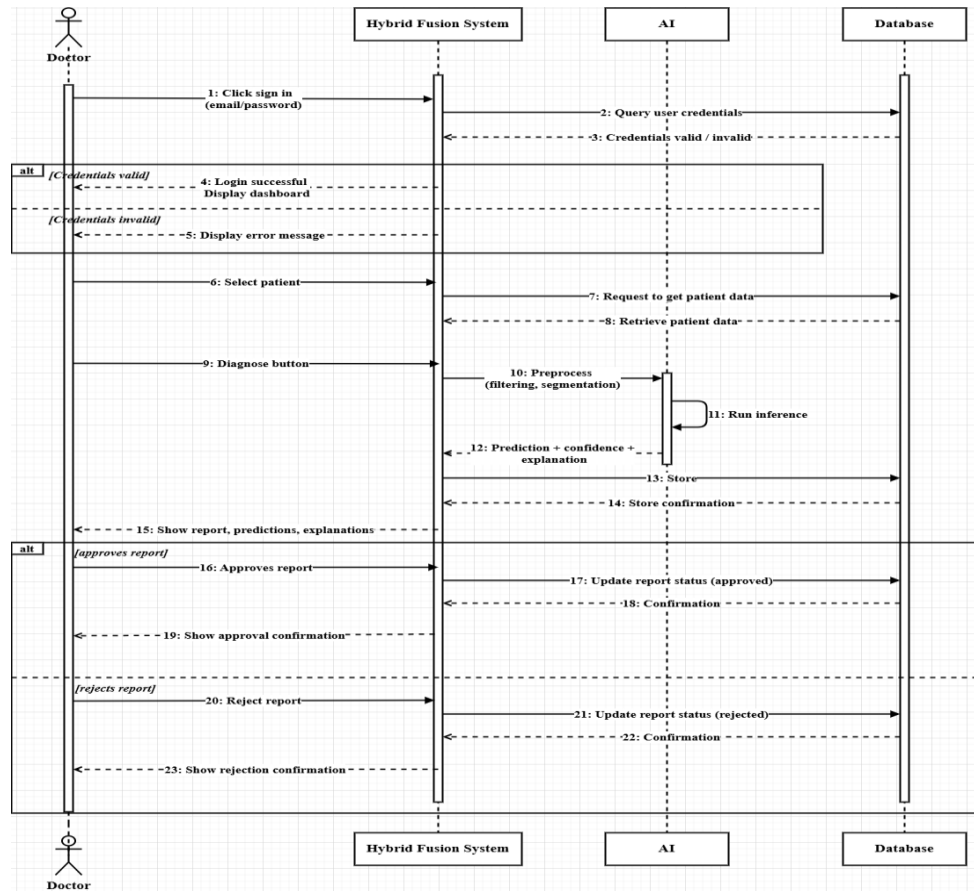


Fig. 4.6: System Sequence Diagram for Doctor

4.3.1.2 Technician System Sequence Diagram

Fig. 4.7 demonstrates the technician system interaction flow, including authentication, patient and doctor selection, ECG upload, validation, and system response handling.

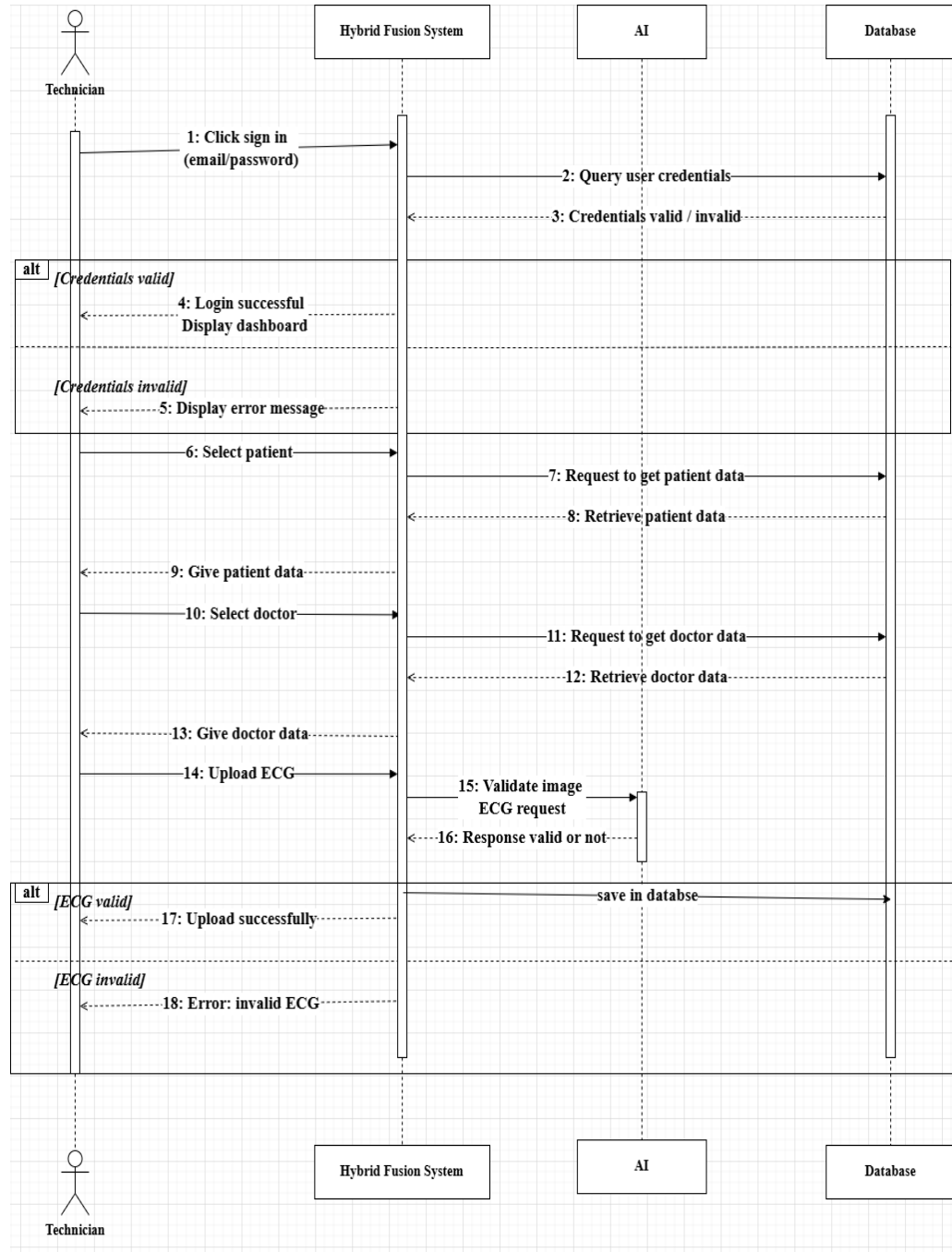


Fig. 4.7: System Sequence Diagram for Technician

4.3.1.3 Patient System Sequence Diagram

Fig. 4.8 shows the patient system interaction sequence, covering authentication, ECG upload, validation, AI-based diagnosis processing, and final result retrieval.

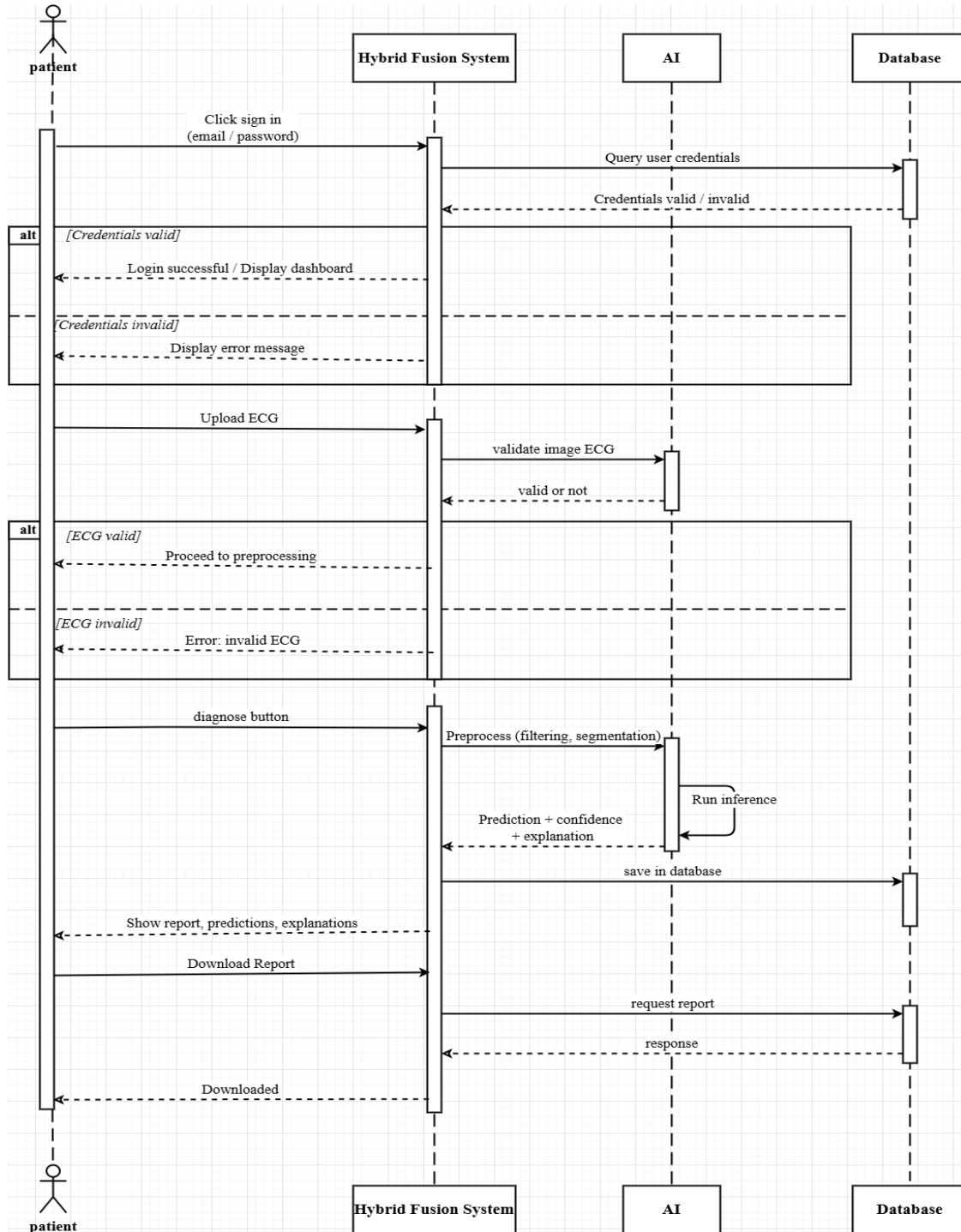


Fig. 4.8: System Sequence Diagram for patient

4.3.2. Decision Tree

A structured flowchart that visualizes decisions and their possible outcomes to support clear, rule-based decision-making.

4.3.2.1. Scenario 1 (Technician Upload & Validate ECG)

The first decision tree in Fig. 4.9 describes the workflow for a technician uploading and validating an ECG.

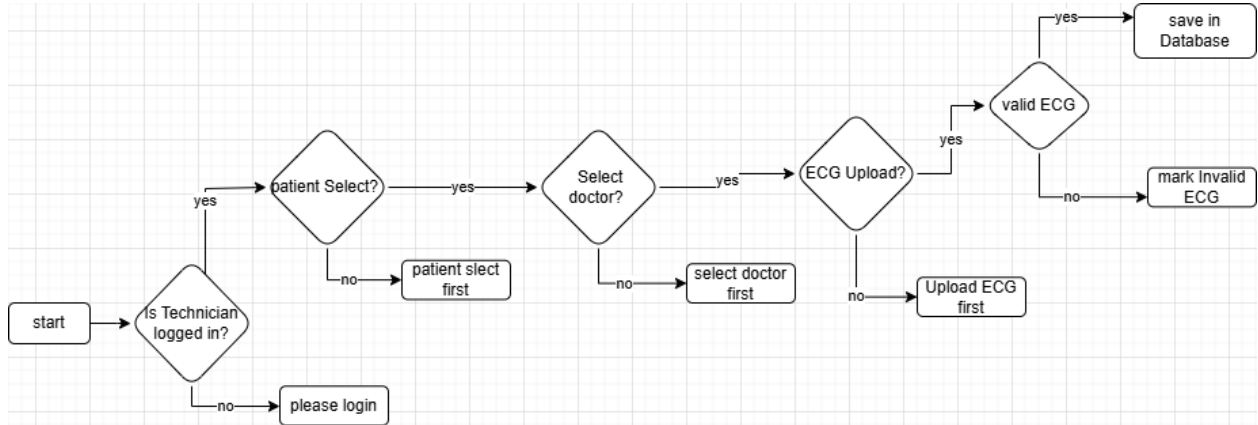


Fig. 4.9: Decision Tree (Technician Upload)

4.3.2.2. Scenario 2 (Doctor Review & Approval)

The second decision tree in Fig. 4.10 illustrates the workflow for the doctor reviewing and approving a patient's report.

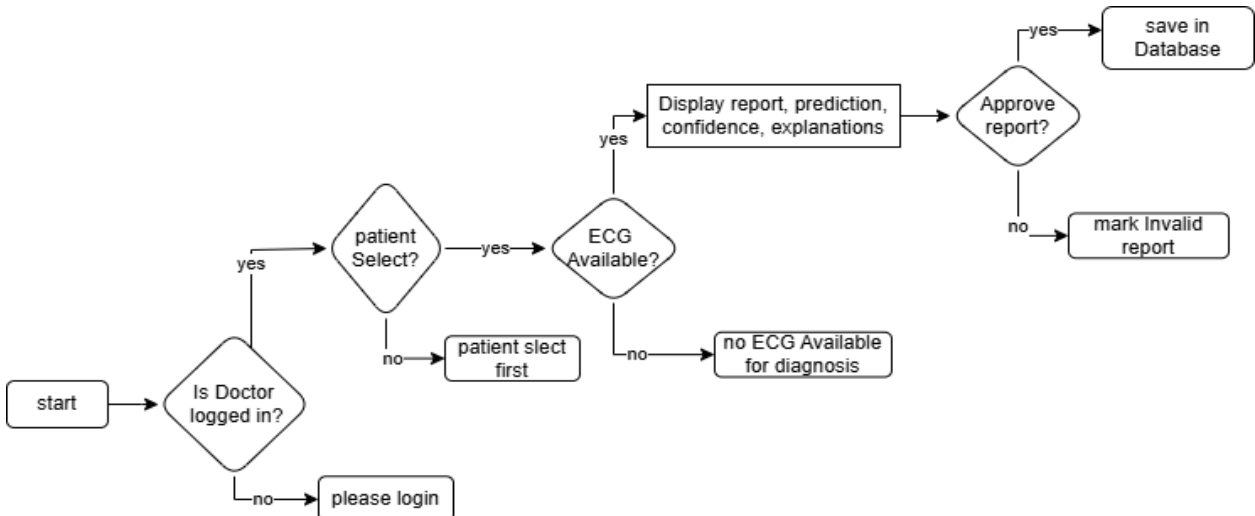


Fig. 4.10: Decision Tree (Doctor Review & Approval)

4.3.2.3. Scenario 3 (Patient View Report)

The second decision tree in Fig. 4.11 represents the workflow for a patient viewing their ECG report.

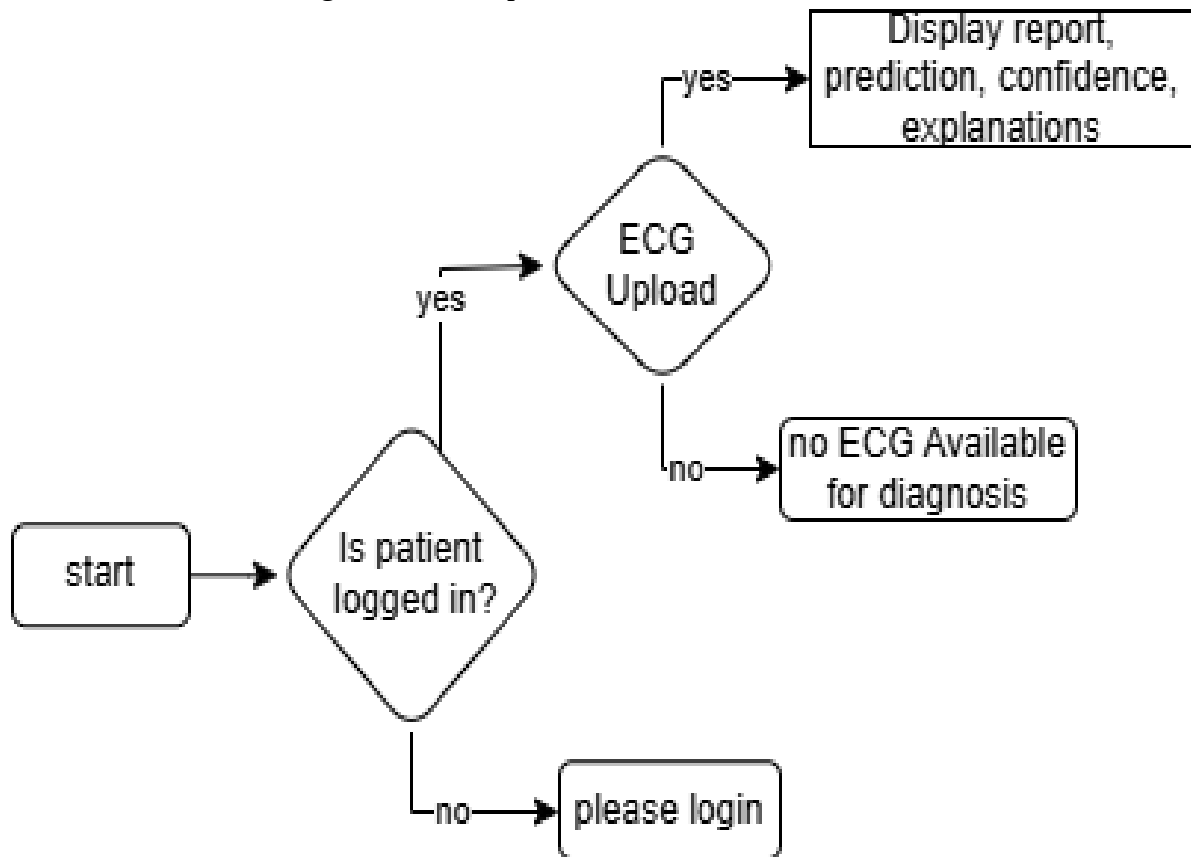


Fig. 4.11: Decision Tree (Patient View Report)

4.3.3. Class Diagram

The Class Diagram Fig. 4.12 provides a structural view of the system by defining its main classes, their attributes, methods, and relationships. It illustrates how users, ECG records, AI modules, and reports interact to form the complete architecture of the Hybrid Multimodal Fusion System. It also highlights the associations and dependencies among classes, ensuring efficient data flow and system coordination. Furthermore, it helps in understanding the logical design and supports implementation and maintenance of the system.

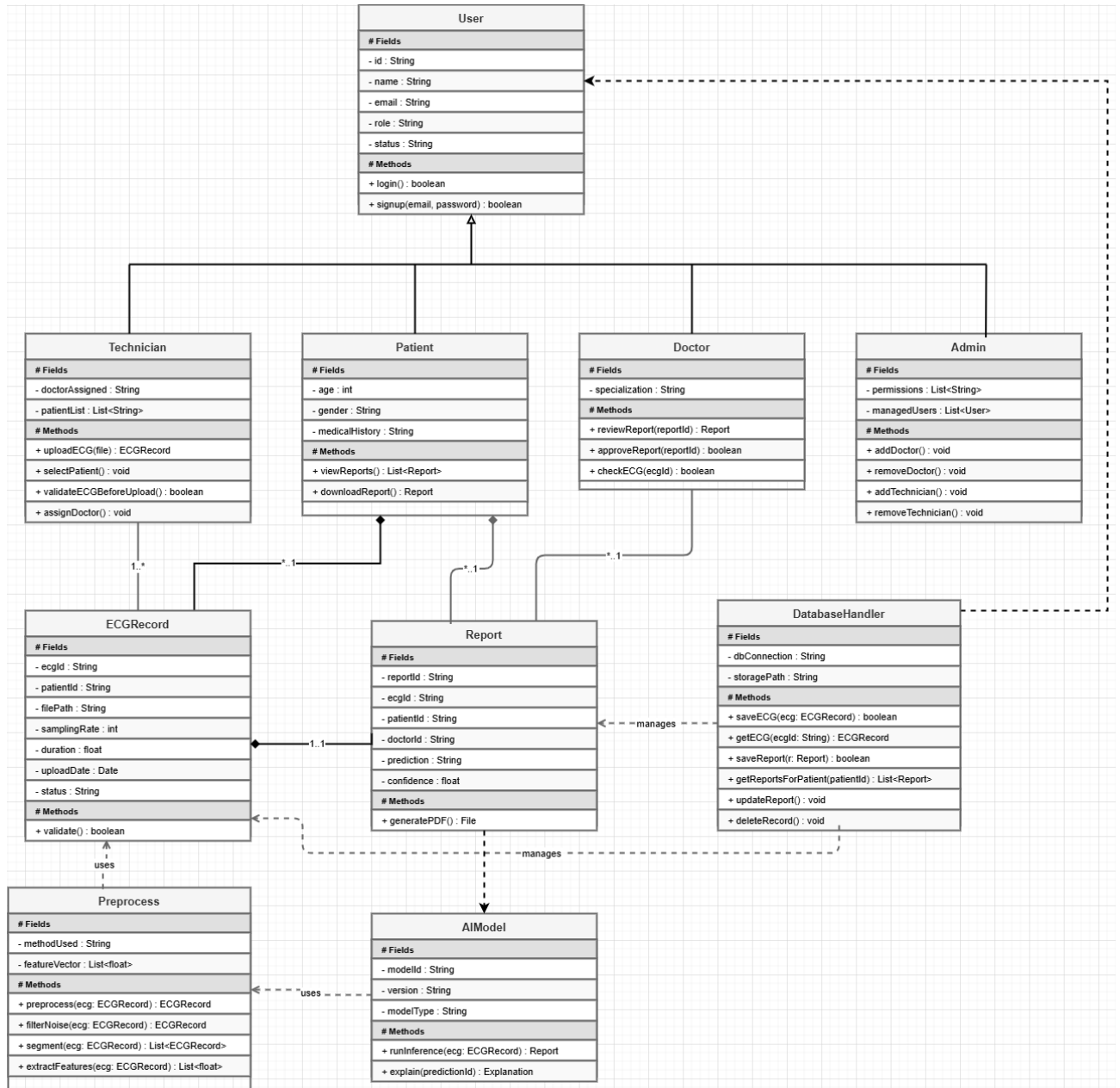


Fig. 4.12: Class diagram

4.3.4. Deployment Diagram

The Deployment Diagram Fig. 4.13 illustrates how the system components are distributed across client, server, and cloud environments, showing the interaction between the web interface, API layer, FastAPI [1] integration with backend, and the AI/ML engine for ECG data processing and prediction. It also highlights the cloud storage for secure data management, efficient communication, and system scalability. Furthermore, it represents the physical architecture of the system, ensuring reliable deployment and performance across different platforms. It helps in understanding how components are hosted, connected, and executed in real-world environments.

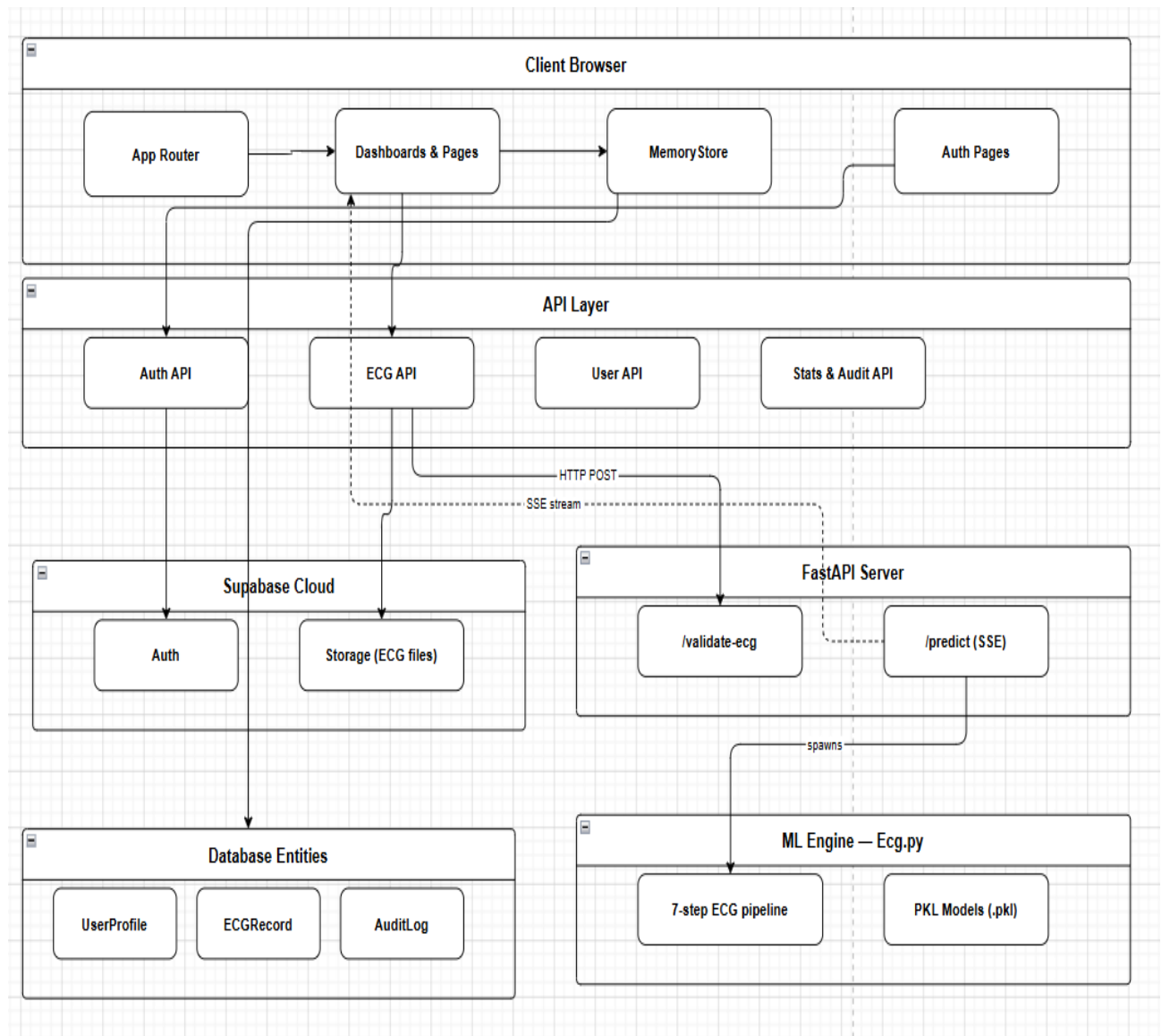


Fig. 4.13: Deployment Diagram

4.3.5. State Transition Diagram

The State Transition Diagram Fig. 4.14 shows how the system transitions between states—from authentication and ECG upload to validation, analysis, review, and report generation—handling both successful flows and error conditions.

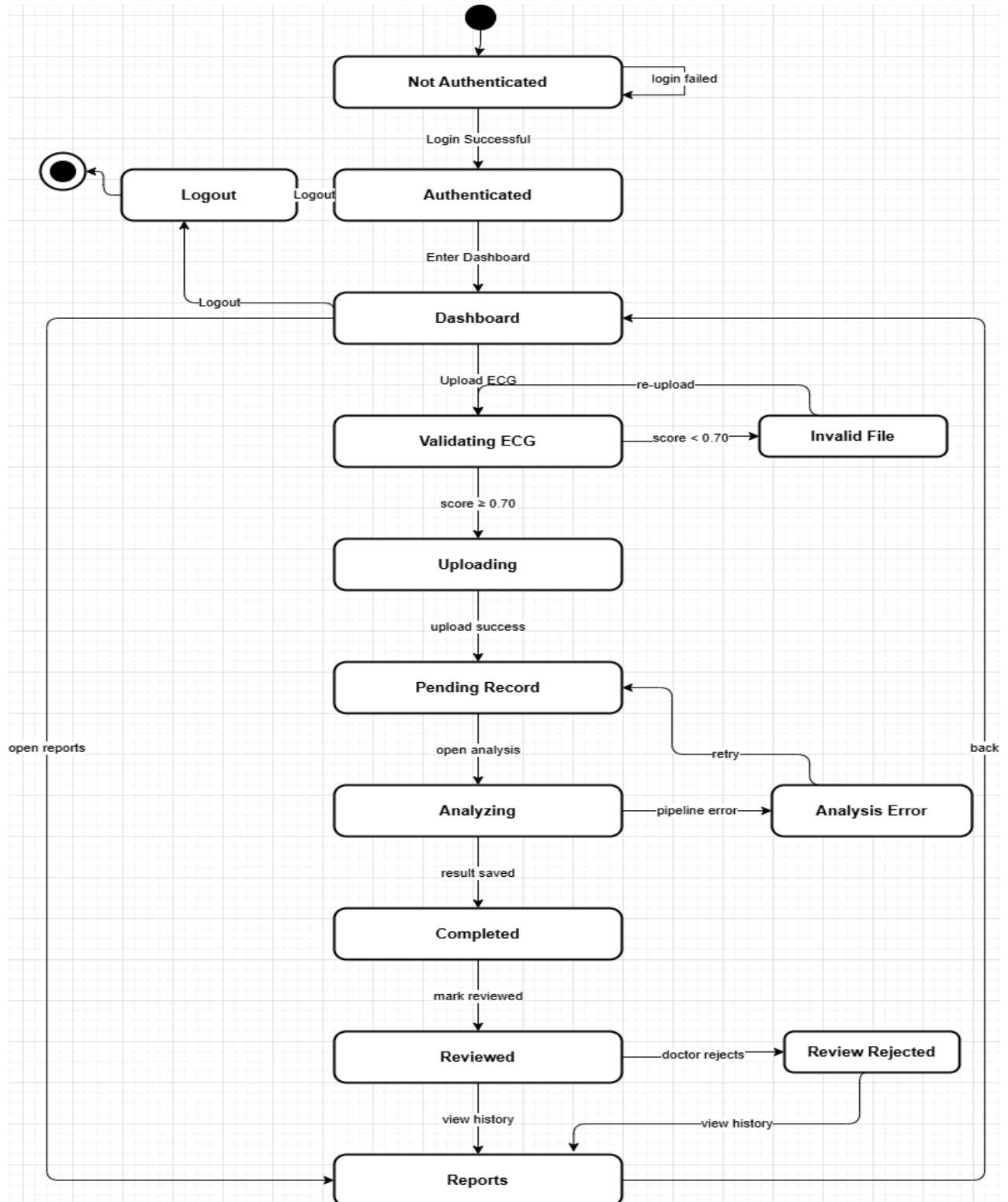


Fig. 4.14: State Transition Diagram

4.3.6. Block Diagram

The Block Diagram illustrates the overall processing pipeline of the cardiovascular disease detection system. As shown in Fig. 4.15, each functional block represents a key stage from ECG image input, lead segmentation, preprocessing, and feature extraction, through to ML classification and final diagnosis output connected sequentially by directional flow arrows.

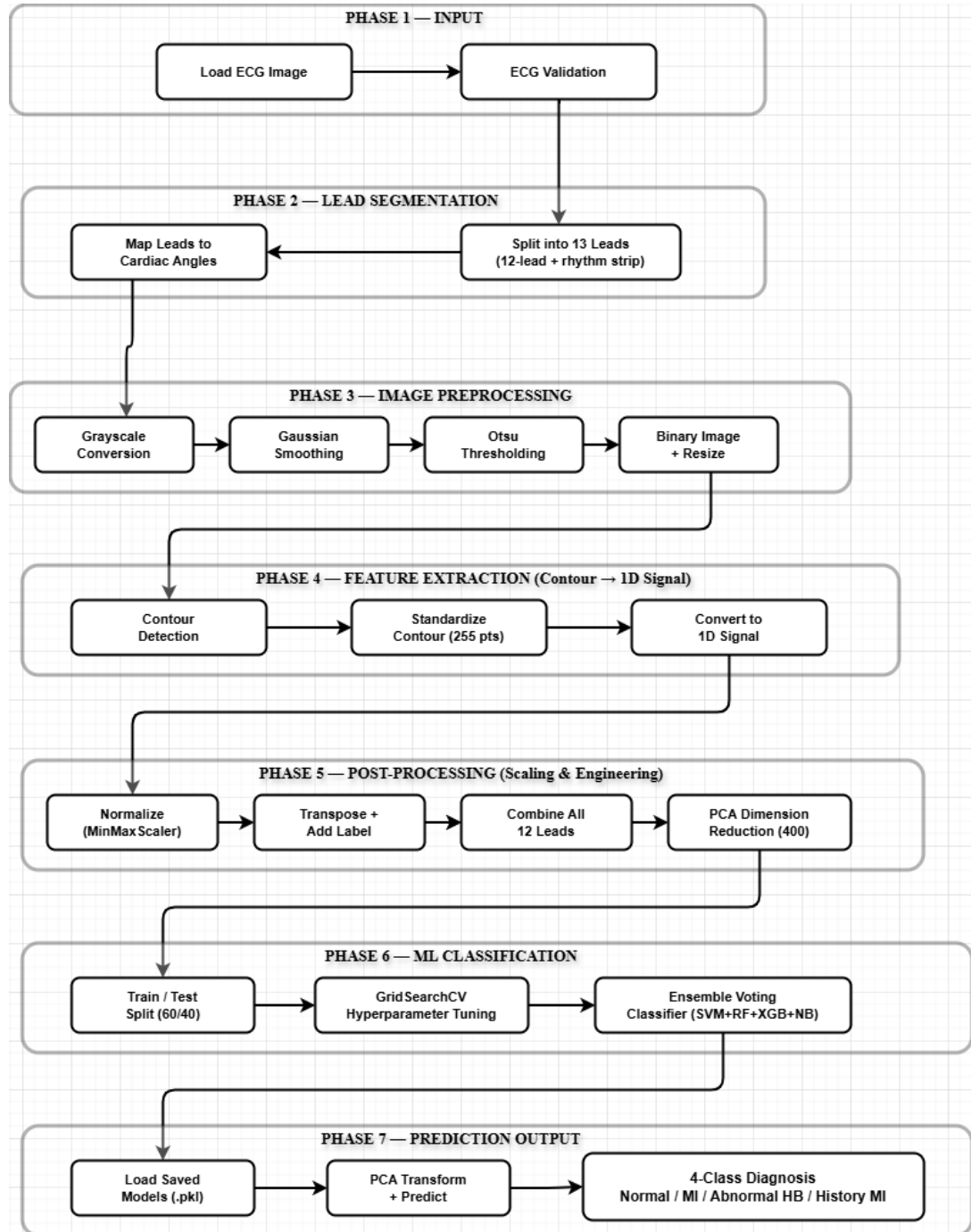


Fig. 4.15: Block Diagram

4.4. Data Representation

The proposed system uses a structured relational database model to manage and organize all core entities related to ECG-based cardiovascular disease detection. PostgreSQL [2] is used to maintain data integrity, enforce entity relationships, and support secure storage of patient records, ECG uploads, AI-generated predictions, reports, and system audit activities. The database schema is designed to ensure clear linkage between users, ECG records, predictions, explanations, and generated reports through well-defined primary and foreign key constraints.

The Entity Relationship Diagram (ERD) visually represents the logical structure of the system database. It defines the cardinality and relationships among key entities such as UserProfile, ECGRecord, Prediction, Report, Explanation, AuditLog, and FileBlob. This design supports efficient data retrieval, traceability, role-based access, and accurate clinical decision support workflows.

Fig. 4.16 illustrates the visual representation of the ERD for the proposed cardiovascular disease detection system.

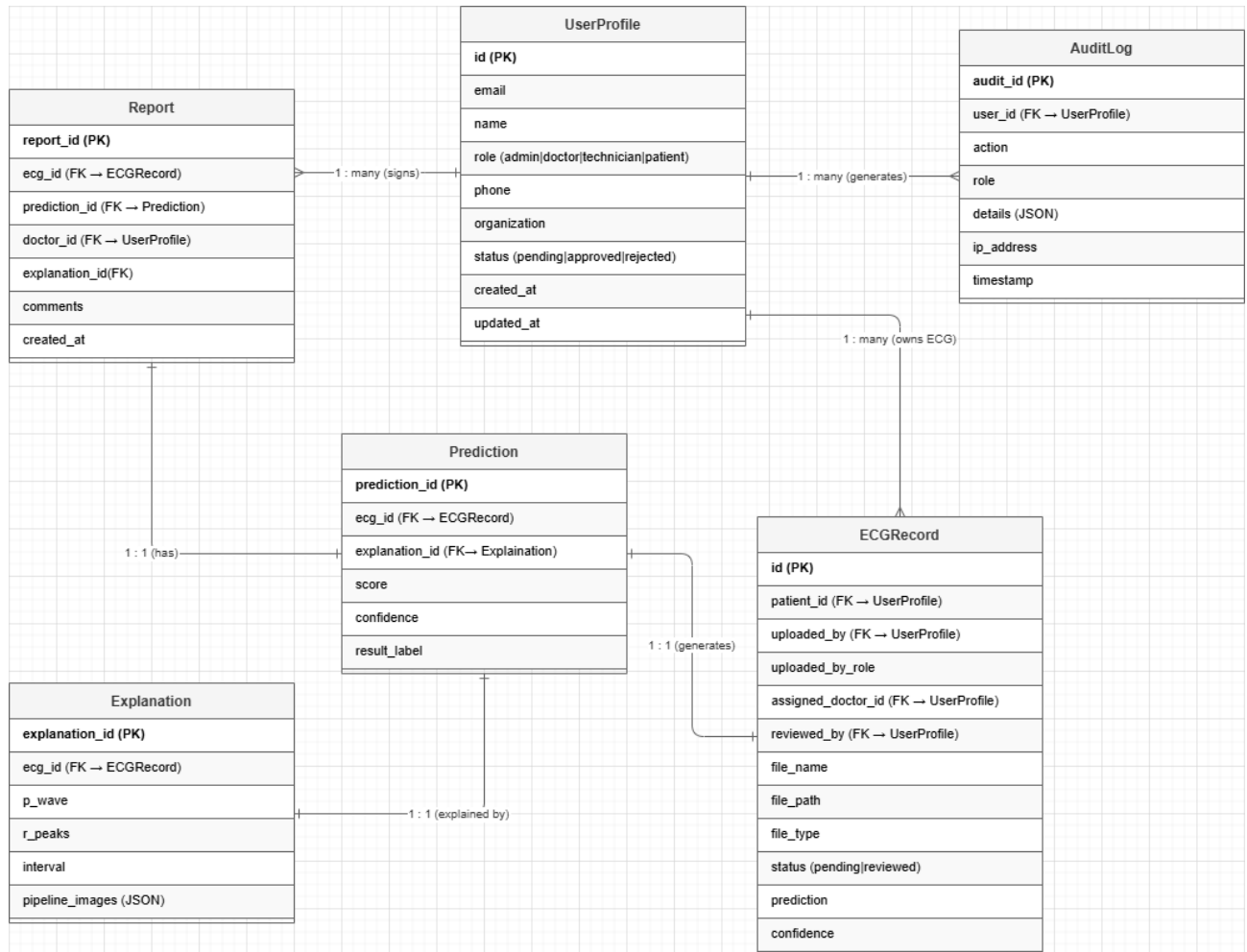


Fig. 4.16: ERD Diagram

5. Implementation

This chapter presents the practical implementation of the Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection. It translates the design models discussed in Chapter 4 into a working web-based clinical decision-support system. The implemented solution supports secure role-based workflows, ECG image validation, lead segmentation, preprocessing, numerical feature generation, dimensionality reduction, machine-learning-based classification, progressive result delivery, and explainable result presentation.

The chapter first summarizes the deployed processing workflow and the major software components. It then describes the algorithms implemented within the ECG processing class and the FastAPI backend. The service interfaces used by the application are documented afterwards. Finally, the user-interface screens are described to demonstrate how Admin, Doctor, Technician, and Patient roles interact with the system. The wording in this chapter is based on the implemented source code and has been written specifically for this project.

The deployed classification pipeline recognizes four output categories: Abnormal Heartbeat, Myocardial Infarction, Normal ECG, and History of Myocardial Infarction. The system is intended to support clinical review; its output does not replace professional medical judgement.

5.1. Implementation Overview

The implemented pipeline begins when an authorized user uploads an ECG image through the web interface. The backend receives the file, creates an isolated temporary working directory, and stores the uploaded bytes for processing. The image is converted into grayscale format and resized to a standard resolution. A structural similarity check is then performed against representative ECG reference images. Only images that satisfy the `conFig.d` acceptance threshold are forwarded to the full analysis pipeline.

After validation, the ECG sheet is divided into twelve standard lead regions and one long-lead region. The standard leads are preprocessed independently using grayscale conversion, Gaussian smoothing, and Otsu thresholding. A contour-based signal extraction procedure converts each standard lead into a normalized one-dimensional representation. The generated lead-level values are combined into a single feature vector, transformed through a previously trained Principal Component Analysis (PCA) model, and submitted to the serialized classifier.

The backend progressively streams intermediate processing updates to the frontend. These updates include the original image, grayscale image, separated leads, preprocessed leads, and extracted contours. Once prediction is complete, the explainability component produces class probabilities, individual model votes, lead-wise measurements, annotated ECG images, lead contribution values, heart-axis estimates, and computer-assisted interpretation notes. Fig. 5.1 illustrates the deployed implementation workflow.

5.2. Implementation Components

The application is implemented as a set of coordinated components. The web layer manages user interaction and role-based workflows, while the Python backend manages ECG

processing and model inference. The ECG class encapsulates the image-processing stages. The FastAPI service coordinates requests, streams progress messages, and returns the final response. The explainability component runs after classification to improve transparency of the generated result. Table 5.1 summarizes the major implementation components.

Table 5.1: Major components used in the implementation

Component	Implementation Layer	Purpose
React with TypeScript	Frontend interface	Provides role-based dashboards, upload forms, progress views, and result presentation.
FastAPI	Backend API service	Receives ECG files, validates requests, coordinates the processing pipeline, and streams progress events [1].
Python	Processing environment	Executes ECG image processing, numerical transformation, model inference, and explainability routines.
scikit-image	Image-processing library	Supports image reading, grayscale conversion, resizing, Gaussian filtering, Otsu thresholding, SSIM calculation, and contour extraction [10].
NumPy and Pandas	Numerical and tabular processing	Support array operations, CSV-based feature storage, and dataframe construction [11], [12].
scikit-learn and Joblib	Machine-learning layer	Support Min-Max scaling, PCA transformation, serialized model loading, and classification [13].
NeuroKit2 and SciPy	Explainability support	Support PQRST estimation, peak analysis, smoothing, and measurement generation [14].
Matplotlib	Visualization layer	Generates separated-lead, preprocessed-lead, contour, and contribution visualizations [15].
Supabase	Application data services	Supports authentication, relational data storage, and file-storage requirements defined in the system architecture [2].

5.3. Algorithms

This section documents the principal algorithms implemented in the application. Each algorithm is described in terms of its objective and operational sequence. The descriptions are expressed at system level so that they remain understandable for future maintainers while accurately reflecting the implemented Python classes and backend service functions.

5.3.1. User Account Management

The user-management workflow controls access to application functions and supports the role-based design described in Chapter 3.

Objective: Create, authenticate, and manage user accounts securely according to assigned system roles.

- **Algorithm Steps:**

- Registration:** Collect the required profile details and credentials from a new user or from an administrator creating a staff account.

- ii. **Profile Storage:** Store the user profile and assigned role in the application database.
- iii. **Credential Verification:** Validate the submitted credentials when the user attempts to log in.
- iv. **Role Identification:** Identify whether the authenticated account belongs to an Admin, Doctor, Technician, or Patient.
- v. **Dashboard Redirection:** Display the dashboard and permitted functions associated with the authenticated role.
- vi. **Session Termination:** Invalidate the active session when the user logs out or when the session expires.

5.3.2. ECG Upload and Preliminary Validation

The preliminary-validation stage prevents unrelated or corrupted files from entering the complete inference pipeline. The backend exposes a dedicated validation endpoint and also repeats the validation step before prediction.

Objective: Verify that an uploaded file follows the expected ECG-sheet structure before initiating computationally intensive processing.

Algorithm Steps:

- i. **Image Acquisition:** Read the uploaded image using the `getImage()` method. If an alpha channel is present, retain only the RGB channels.
- ii. **Grayscale Conversion:** Convert the input image into grayscale format using `GrayImgae()` and resize it to 1572 x 2213 pixels.
- iii. **Reference Loading:** Load four representative ECG reference images corresponding to abnormal heartbeat, myocardial infarction, normal ECG, and history of myocardial infarction.
- iv. **Structural Comparison:** Calculate the Structural Similarity Index Measure (SSIM) between the uploaded grayscale image and each reference image.
- v. **Score Selection:** Select the maximum similarity score obtained from the four comparisons.
- vi. **Acceptance Decision:** Accept the image when the maximum similarity score is at least 0.70; otherwise, return a validation error and request a valid ECG image.

The preliminary-validation logic is implemented through the methods `getImage()`, `GrayImgae()`, and `CheckSimilarity()` in the ECG class. The FastAPI service exposes this check through the `POST /validate-ecg` endpoint.

5.3.3. ECG Lead Segmentation

After successful validation, the complete ECG sheet is divided into fixed regions that correspond to the standard lead arrangement used by the selected dataset. The layout contains twelve standard leads and one long-lead region.

- **Objective:** Separate the ECG sheet into lead-specific images so that each

standard lead can be processed independently.

- **Algorithm Steps:**

- Input Retrieval:** Receive the validated ECG image from the preliminary-validation stage.
- Fixed-Region Cropping:** Crop the image using predefined row and column boundaries associated with the expected ECG-sheet layout.
- Standard-Lead Collection:** Create twelve lead images representing Lead I, aVR, V1, V4, Lead II, aVL, V2, V5, Lead III, aVF, V3, and V6.
- Long-Lead Collection:** Extract the additional long-lead strip for visual review and preprocessing.
- Lead Packaging:** Store all thirteen regions in an ordered list and return the list to the backend orchestration function.
- Visualization:** Generate separate Fig.s for the twelve standard leads and the long-lead strip.

Table 5.2 presents the lead arrangement extracted by the DividingLeads() method.

Table 5.2: Lead arrangement extracted by DividingLeads()

ECG sheet region	Column 1	Column 2	Column 3	Column 4
First row	Lead I	aVR	V1	V4
Second row	Lead II	aVL	V2	V5
Third row	Lead III	aVF	V3	V6
Additional strip	Long lead	-	-	-

DividingLeads() method returns thirteen image regions. The twelve standard leads are used for classifier feature generation, whereas the additional long-lead region is retained for visualization and review.

5.3.4. Lead-Wise Image Preprocessing

Each segmented lead is processed independently to improve the separation between the waveform trace and the background. This step creates binary lead images suitable for subsequent contour extraction.

- **Objective:** Standardize the segmented lead images and enhance the visibility of waveform structures before numerical feature extraction.
- **Algorithm Steps:**
 - Grayscale Conversion:** Convert each cropped standard-lead image into grayscale format.
 - Gaussian Smoothing:** Apply Gaussian filtering with sigma equal to 1 to reduce minor image noise.
 - Threshold Estimation:** Compute an Otsu threshold value for each blurred lead image.

- iv. **Binary Conversion:** Create a binary image by identifying pixels darker than the calculated threshold.
- v. **Standard Resizing:** Resize each standard-lead binary image to 300 x 450 pixels.
- vi. **Output Generation:** Save the preprocessed twelve-lead Fig. and the preprocessed long-lead Fig. for interface presentation.

The preprocessing routine is implemented through *PreprocessingLeads()*. The operation is also reproduced within the FastAPI orchestration function so that the frontend can receive each processed lead as an intermediate visualization.

5.3.5. Contour-Based Signal Extraction and Scaling

The classifier requires a consistent numerical representation rather than raw image pixels. Therefore, the preprocessed lead images are converted into normalized one-dimensional values through contour extraction and scaling.

- **Objective:** Transform each standard ECG lead image into a normalized one-dimensional numerical signal for model inference.
- **Algorithm Steps:**
 - i. **Lead Conversion:** Convert the current lead image into grayscale format.
 - ii. **Noise Smoothing:** Apply Gaussian filtering with sigma equal to 0.7.
 - iii. **Binary Mapping:** Calculate the Otsu threshold and construct a binary lead image.
 - iv. **Standardization:** Resize the binary image to 300 x 450 pixels.
 - v. **Contour Identification:** Apply contour detection at the `conFig.d` level of 0.8.
 - vi. **Contour Selection:** Select the largest detected contour as the primary representation of the waveform structure.
 - vii. **Fixed-Length Conversion:** Resize the selected contour to 255 points with two coordinate values per point.
 - viii. **Min-Max Scaling:** Normalize the selected contour values using `MinMaxScaler`.
 - ix. **Lead-Level Storage:** Store the scaled one-dimensional values in the corresponding lead-level CSV file.

The complete routine is implemented in *SignalExtraction_Scaling()*. The generated contour Fig.s are also streamed to the web interface so that users can observe the image-to-signal conversion stage.

5.3.6. Feature Consolidation and PCA Transformation

The normalized lead-level values are combined before model inference. Dimensionality reduction is then applied using the PCA transformation learned during model training.

- **Objective:** Construct the classifier input vector and reduce its dimensionality while retaining the learned feature representation.
- **Algorithm Steps:**
 - i. **Lead File Retrieval:** Read the scaled one-dimensional CSV file generated for the first ECG lead.
 - ii. **Horizontal Concatenation:** Read the remaining scaled lead files and concatenate their values horizontally to construct a single dataframe.
 - iii. **PCA Model Loading:** Load the serialized PCA model from PCA_ECG (1).pkl using Joblib.
 - iv. **Feature Transformation:** Apply the transform() operation of the loaded PCA model to the consolidated dataframe.
 - v. **Output Construction:** Convert the reduced feature representation into a dataframe for classifier inference.

Feature consolidation and dimensionality reduction are implemented through *CombineConvert1Dsignal()* and *DimensionalReduciton()*, respectively. PCA is used as a linear dimensionality-reduction technique within scikit-learn [10].

5.3.7. Cardiovascular Disease Classification

The reduced feature dataframe is passed to the trained classifier. The serialized model returns a numeric class label, which is mapped to a human-readable diagnostic category.

- **Objective:** Generate an AI-assisted cardiovascular disease classification from the processed ECG image.
- **Algorithm Steps:**
 - i. **Classifier Loading:** Load the serialized model stored as Heart_Disease_Prediction_using_ECG (4).pkl.
 - ii. **Inference:** Submit the PCA-transformed feature dataframe to the model predict() operation.
 - iii. **Class Mapping:** Convert the numeric result into the corresponding cardiovascular disease category.
 - iv. **Confidence Retrieval:** When supported by the loaded estimator, calculate and normalize the prediction probabilities and select the highest value as the displayed confidence score.
 - v. **Result Packaging:** Attach the predicted label, similarity score, confidence value, and explainability output to the final API response.

Table 5.3 summarizes the cardiovascular disease class mapping used by the ModelLoad_predict() function. The predicted class codes are mapped to their corresponding diagnostic labels before being displayed to the user.

Table 5.3: Cardiovascular disease class mapping used by `ModelLoad_predict()`

Class code	Displayed classification
0	Abnormal Heartbeat
1	Myocardial Infarction
2	Normal ECG
3	History of Myocardial Infarction

5.3.8. FastAPI Pipeline Orchestration and Progressive Streaming

The backend service coordinates the complete inference sequence and sends progress information to the frontend while analysis is running. This design improves user awareness during stages that require additional processing time.

- **Objective:** Execute ECG analysis in an isolated working environment and deliver intermediate progress updates without blocking the web interface.
- **Algorithm Steps:**
 - i. **Request Reception:** Receive ECG image bytes through a FastAPI file-upload endpoint.
 - ii. **Temporary Workspace:** Create a dedicated temporary directory for the current analysis request.
 - iii. **Resource Preparation:** Copy the serialized model files and similarity-reference images into the temporary workspace.
 - iv. **Background Execution:** Run the processing function through a `ProcessPoolExecutor` so that computational work does not block the main API event loop.
 - v. **Queue Messaging:** Place step number, progress value, status message, and intermediate image data into a multiprocessing queue.
 - vi. **Base64 Encoding:** Convert generated images into base64-encoded data URLs for delivery to the frontend.
 - vii. **Event Streaming:** Yield queue messages through a `StreamingResponse` with the text/event-stream media type.
 - viii. **Finalization:** Return the final result after processing and remove the temporary workspace.

The backend streams updates for initialization, structural-similarity analysis, lead segmentation, Otsu thresholding, contour generation, PCA encoding, classifier execution, explainability generation, PQRST analysis, and report completion. Server-sent event streams allow servers to push incremental updates to web clients over an HTTP connection [16].

5.3.9. Explainability Analysis and Clinical Interpretation

Explainability processing is executed after the primary classifier has generated its

output. Its purpose is to provide supporting evidence and interpretable summaries without changing the classifier input or the predicted class.

- **Objective:** Present transparent, computer-assisted evidence that helps a medical user review the generated cardiovascular disease classification.
- **Algorithm Steps:**
 - i. **Signal Retrieval:** Read the twelve scaled one-dimensional lead CSV files generated during the primary inference pipeline.
 - ii. **PQRST Estimation:** Detect ECG waveform landmarks using NeuroKit2 ECG cleaning, peak detection, and wave delineation operations. When required, apply a SciPy-based fallback routine [14].
 - iii. **Measurement Generation:** Calculate lead-level estimates such as heart rate, rhythm regularity, amplitudes, PR interval, QRS duration, QT interval, ST deviation, morphology indicators, and signal-quality values.
 - iv. **Annotated Visualization:** Generate lead images or waveform plots with P, Q, R, S, and T markers and supporting measurement labels.
 - v. **Model Breakdown:** Retrieve normalized class probabilities and, where available, individual estimator votes from the ensemble classifier.
 - vi. **Lead Contribution Estimation:** Map Random Forest feature importance values through the PCA components and group them by lead to estimate relative contribution percentages.
 - vii. **Axis Estimation:** Estimate the frontal-plane heart-axis category using the derived measurements for Lead I and aVF.
 - viii. **Interpretation Notes:** Generate structured interpretation notes and lead-level reports for interface presentation.
 - ix. **Clinical Disclaimer:** Display the generated findings as computer-assisted estimates that require verification by a qualified physician.

The main orchestration function for this stage is *run_full_explainability()*. It returns class probabilities, individual votes, lead measurements, lead reports, lead-importance information, axis information, clinical interpretation notes, and annotated signals.

5.3.10. Report Review and Workflow Completion

After the backend returns the prediction and explainability information, the application presents the result through role-specific workflows. A technician can submit an ECG record for analysis, a doctor can review the result and approve or reject the report, and a patient can access an approved report. Administrative users can review system activity and manage staff accounts according to the requirements defined in Chapter 3.

- **Objective:** Complete the clinical workflow by presenting the AI-assisted result for controlled review, approval, and report access.
- **Algorithm Steps:**
 - i. **Result Display:** Present the generated classification, confidence value, similarity score, and available visual evidence.
 - ii. **Doctor Review:** Allow the doctor to inspect the ECG analysis and explainability information.
 - iii. **Decision Recording:** Record the doctor approval or rejection decision for the generated report.
 - iv. **Patient Access:** Make the approved report available to the associated patient.
 - v. **Audit Support:** Maintain traceable system events for administrative review.

5.4. APIs and Service Interfaces

The system uses API-based communication between the web interface and backend services. The ECG analysis itself is performed locally by the deployed Python pipeline; no third-party diagnostic API is used to produce the classification. FastAPI is used to receive uploaded files and return results [1]. Supabase services support the authentication and application-data requirements described in the system architecture [2]. Table 5.4 summarizes the relevant interfaces.

Table 5.4: APIs and service interfaces used by the system

API or interface	Category	Purpose of usage	Function or integration point
Supabase Authentication Service	External application service	Supports secure account registration, login, session handling, and role-based access for Admin, Doctor, Technician, and Patient accounts.	Integrated through the frontend authentication service layer.
Supabase PostgreSQL Service	External application service	Stores structured records such as user profiles, ECG records, predictions, reports, explanations, and audit events.	Integrated through the application data-access layer.
Supabase Storage Service	External application service	Stores uploaded ECG files and report-related file objects according to the system architecture.	Integrated through the application storage layer.
POST /validate-ecg	FastAPI endpoint	Performs preliminary image validation and returns whether the submitted file matches the expected ECG-sheet structure.	validate_ecg() and __validate_ecg_sync()
POST /predict	FastAPI endpoint	Executes the full ECG processing and prediction workflow and streams progress messages to the frontend.	predict() event_generator() process_ecg_pipeline_queue()
text/event-stream	Streaming interface	Transfers progressive server-sent status events and partial images during analysis.	StreamingResponse() with media_type="text/event-stream"

5.5. User Interface

The web interface is designed to present a clear and role-oriented workflow. Each user sees only the functions relevant to their responsibilities. The following subsections describe the interface screens.

5.5.1. User Registration Screen

The registration screen allows a new user to provide the required account details as shown in Fig. 5.1. Patient accounts may be registered through the public workflow, whereas staff-related accounts are managed according to administrative approval rules.

Create Account
Register for CardioDetect AI

Admin already exists — request an admin invite if needed.

Full Name *
john doe

Email *
user@example.com
Email will be auto-lowercased on submit

Role *
Patient ▼

Contact *
+92XXXXXXXXXX
Use Pakistani E.164 format: +92 followed by 10 digits

Password *
Create a strong password
8+ chars including uppercase, lowercase, number & special character

Confirm Password *
Re-enter password

Organization (optional)
Hospital or Clinic

Already have an account? [Sign in](#)

Create Account

Fig. 5.1: User registration screen interface

5.5.2. User Login Screen

The login screen allows registered users to enter their credentials securely as shown in Fig. 5.2. After successful authentication, the system redirects the user to the dashboard associated with the assigned role.

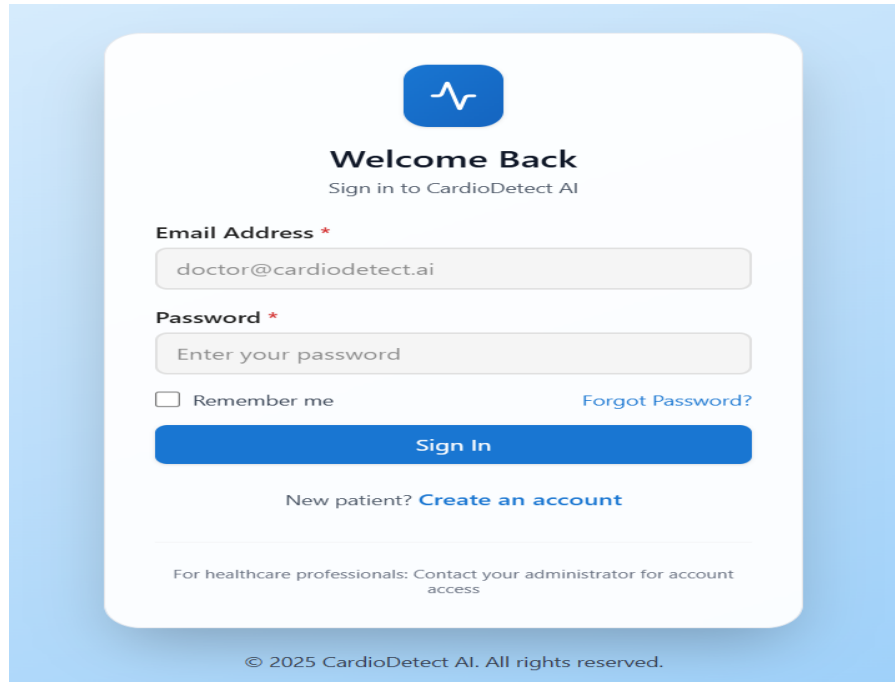


Fig. 5.2: User login screen interface

5.5.3. Doctor Dashboard

The doctor dashboard, as shown in Fig. 5.3, provides access to assigned patient records, ECG analysis results, pending reports, and report-review functions. It supports the doctor in examining AI-assisted findings before approval or rejection.

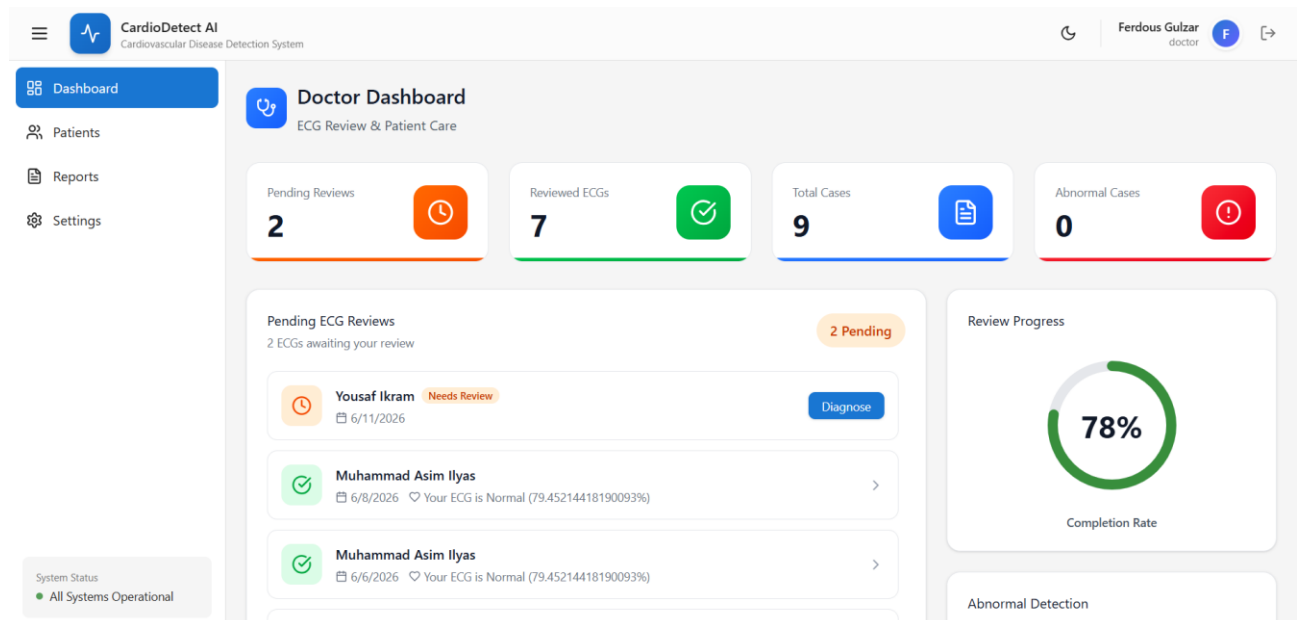


Fig. 5.3: Doctor Dashboard Interface

5.5.4. Technician Dashboard

The technician dashboard, as shown in Fig. 5.4, supports operational tasks such as patient selection, doctor selection, ECG image upload, validation, and submission for analysis.

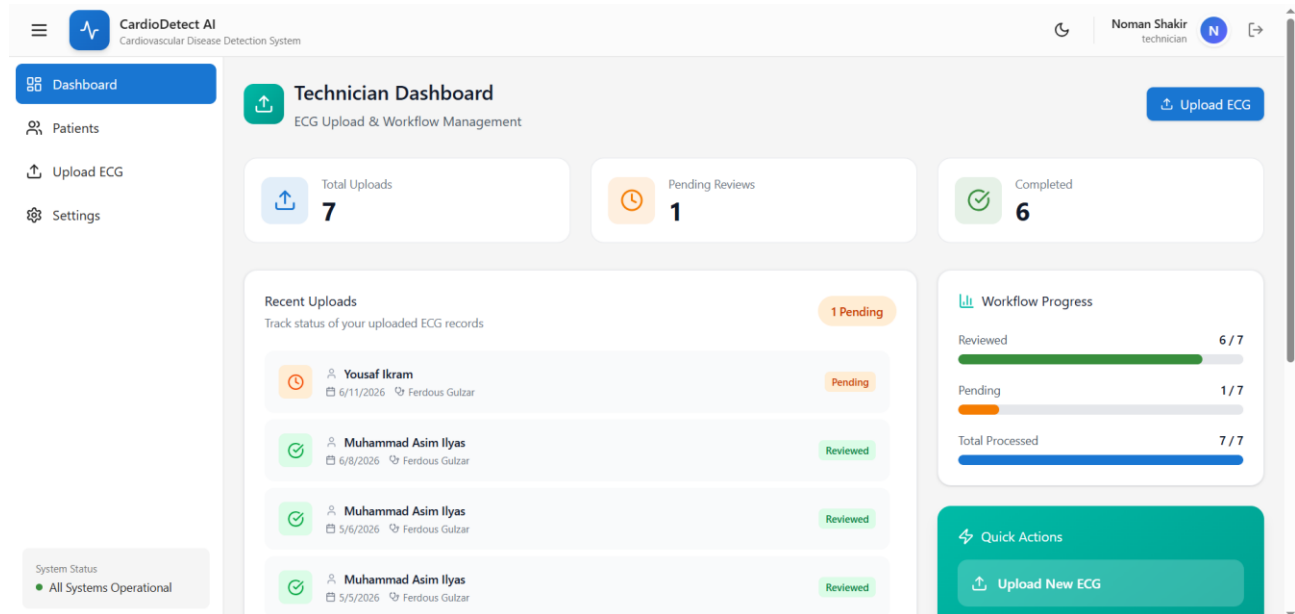


Fig. 5.4: Technician Dashboard Interface

5.5.5. Patient Dashboard

The patient dashboard, as shown in Fig. 5.5, provides controlled access to available ECG reports and related account functions. Only approved reports should be presented through the patient workflow.

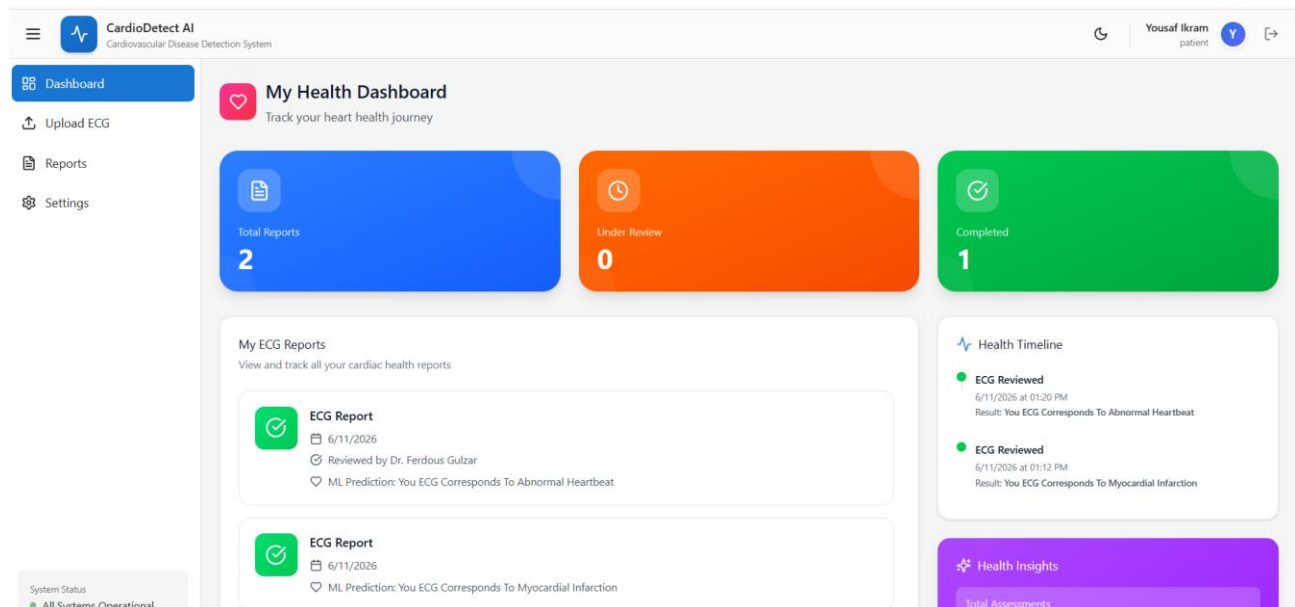


Fig. 5.5: Patient dashboard interface

5.5.6. Admin Dashboard

The admin dashboard, as shown in Fig. 5.6, provides an overview of user and system-management functions. It enables administrative users to manage staff accounts and review audit-related information.

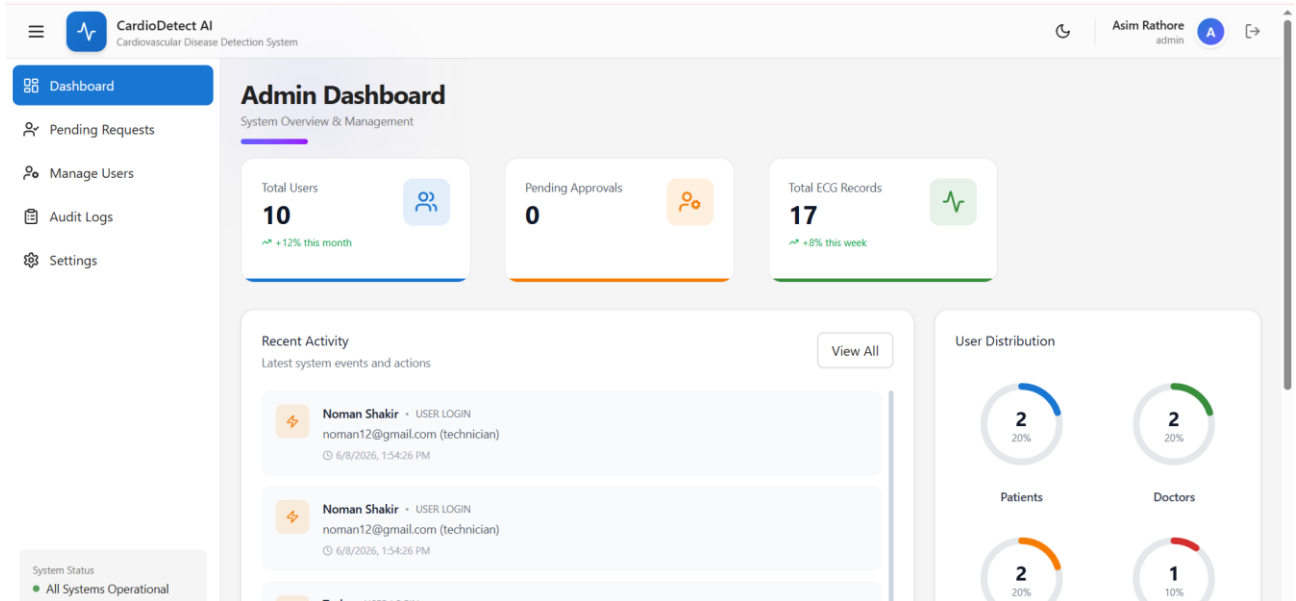


Fig. 5.6: Admin dashboard interface

5.5.7. Patient Selection Screen

The patient-selection screen, as shown in Fig. 5.7, allows an authorized user to search for and select the patient record associated with an ECG upload or review workflow.

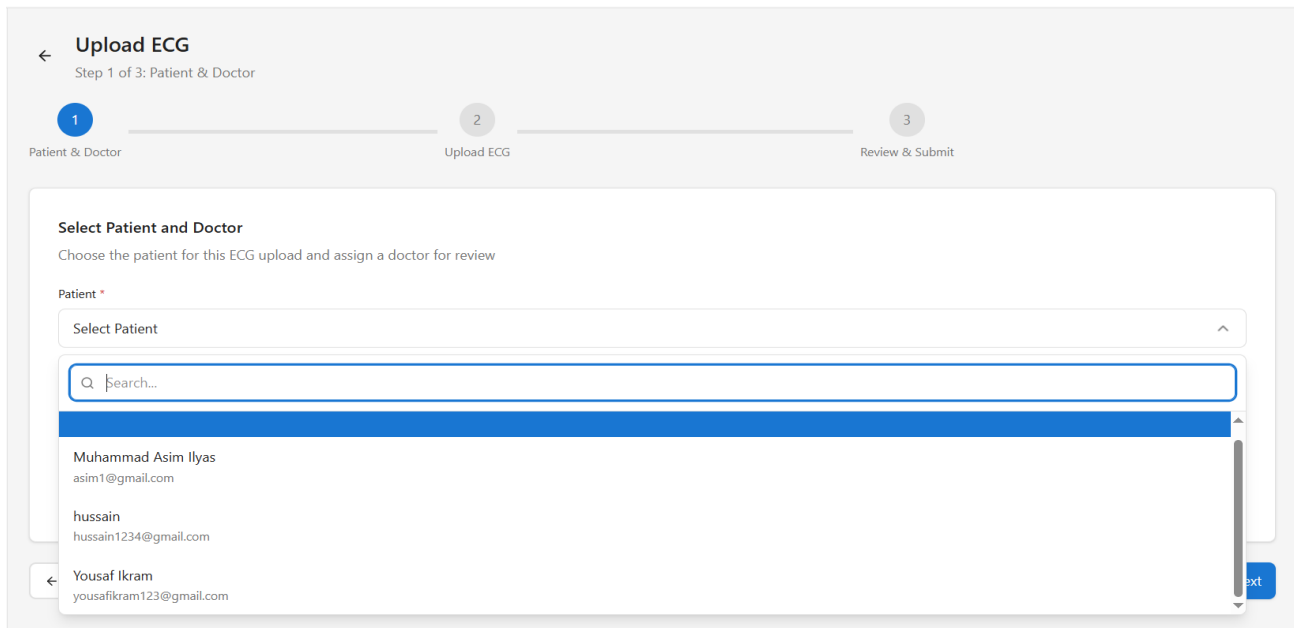


Fig. 5.7: Patient selection screen interface

5.5.8. Doctor Selection Screen

The doctor-selection screen, as shown in Fig. 5.8, allows the relevant workflow user to identify the doctor responsible for reviewing the ECG analysis report.

Fig. 5.8: Doctor Selection screen interface

5.5.9. ECG Upload Screen

The ECG upload screen, as shown in Fig. 5.9, provides a file-selection area for submitting an ECG image. The interface should guide the user to select a suitable ECG sheet before analysis begins.

Fig. 5.9: ECG upload screen interface

5.5.10. ECG Validation Screen

The ECG validation screen, as shown in Fig. 5.10, communicates whether the uploaded image satisfies the expected dataset structure. Invalid or corrupted files are rejected before full processing.

Accepted formats: PNG, JPG, JPEG (Max 10MB)

Validating ECG Image Structure...

Review & Submit

Please review all information before submitting

Patient Information

Patient: Yousaf Ikram

Assigned Doctor

Doctor: Ferdous Gulzar

File Information

File Name: HB(37).jpg
File Size: 0.67 MB
File Type: JPEG

✓ Ready to submit! The ECG will be sent to the assigned doctor for review and diagnosis.

← Back Submit for Review

Fig. 5.10: ECG validation screen interface

5.5.11. ECG Analysis Progress Screen

The analysis-progress screen, as shown in Fig. 5.11, displays the current pipeline stage and completion percentage. Intermediate updates are delivered through the streaming backend response.

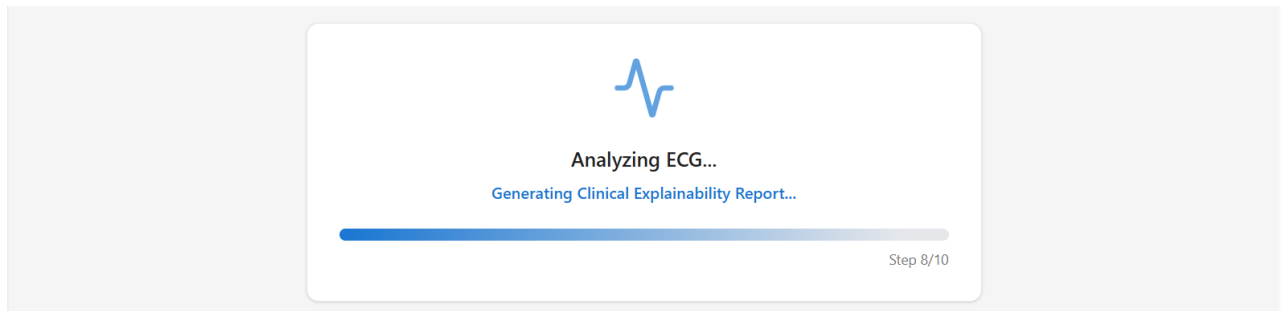


Fig. 5.11: ECG analysis progress screen interface

5.5.12. ECG Processing Visualization Screen

The processing-visualization screen, as shown in Fig. 5.12, presents generated intermediate images, including the original image, grayscale image, separated leads, preprocessed leads, and extracted contours.

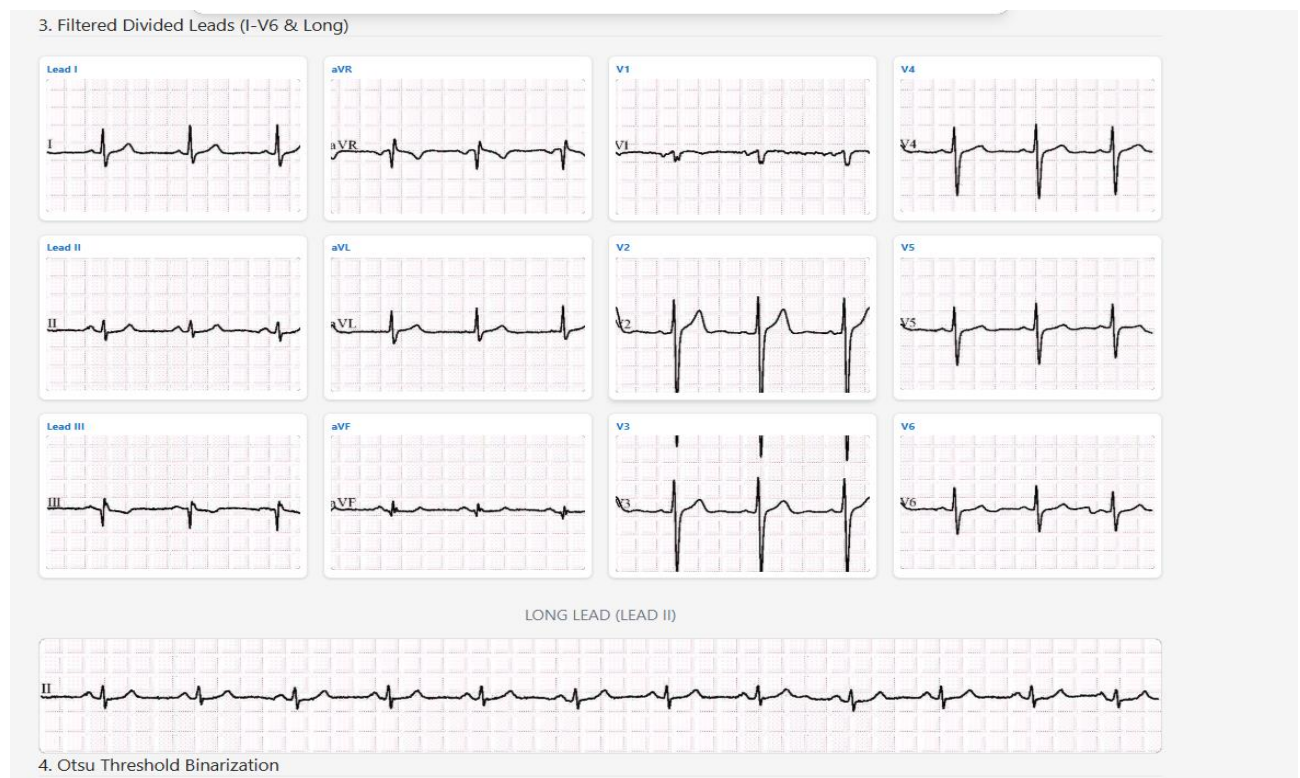


Fig. 5.12: ECG processing visualization interface (Segmented leads)

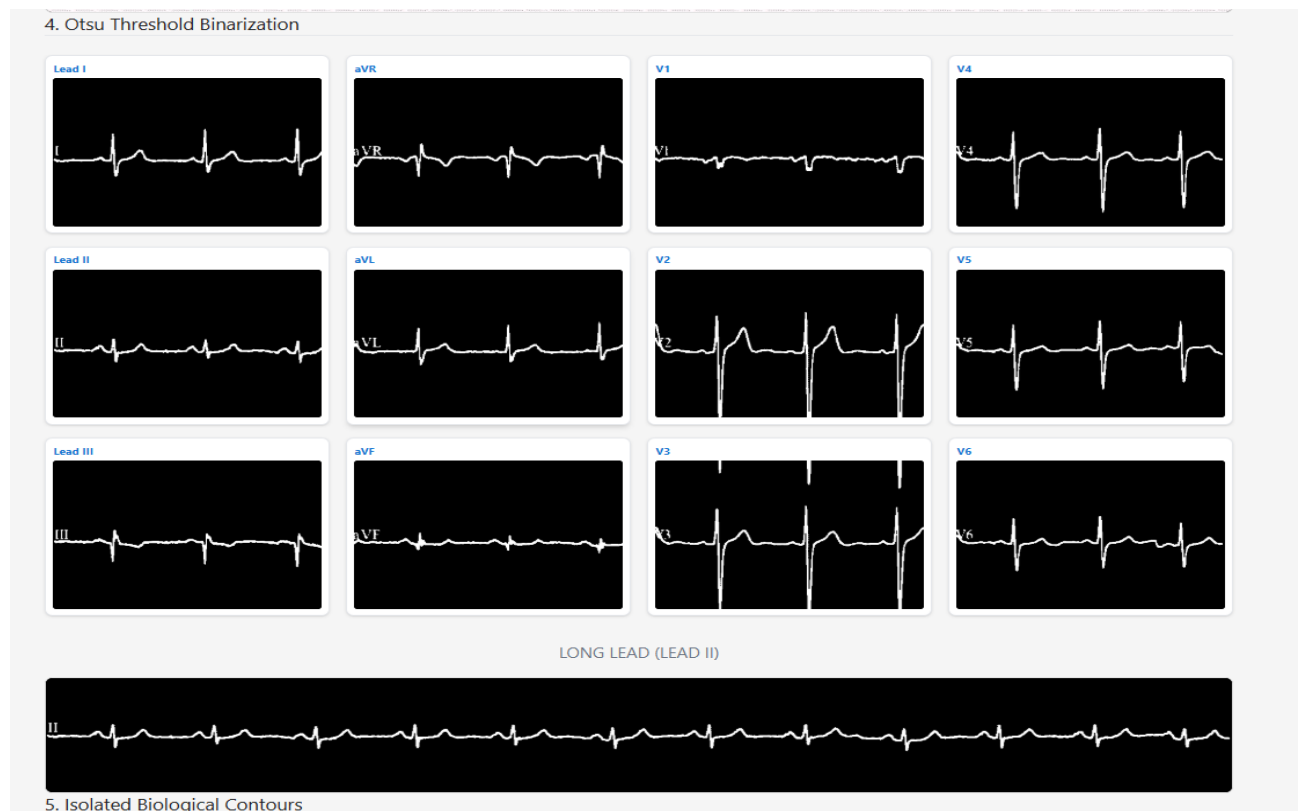


Fig. 5.13: ECG processing visualization interface (Binary lead images)



Fig. 5.14: ECG processing visualization interface(Counters Images)

5.5.13. Disease Prediction Result Screen

The prediction-result, as shown in Fig. 5.15, screen displays the generated cardiovascular disease class, confidence value, and structural-similarity score.

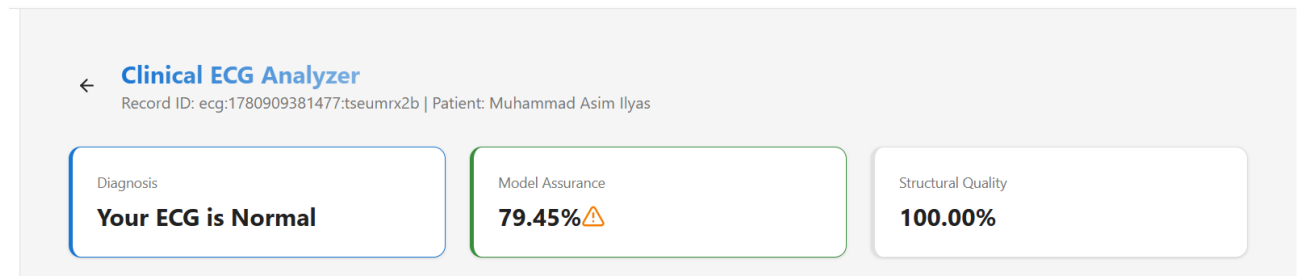


Fig. 5.15: Cardiovascular disease prediction interface

5.5.14. Explainability Analysis Screen

The explainability screen, as shown in Fig. 5.16, presents class probabilities, individual model votes, lead-wise measurements, annotated ECG leads, lead-contribution information, axis estimates, and clinical interpretation notes.



Fig. 5.16: Explainability analysis interface (annotated ECG leads)

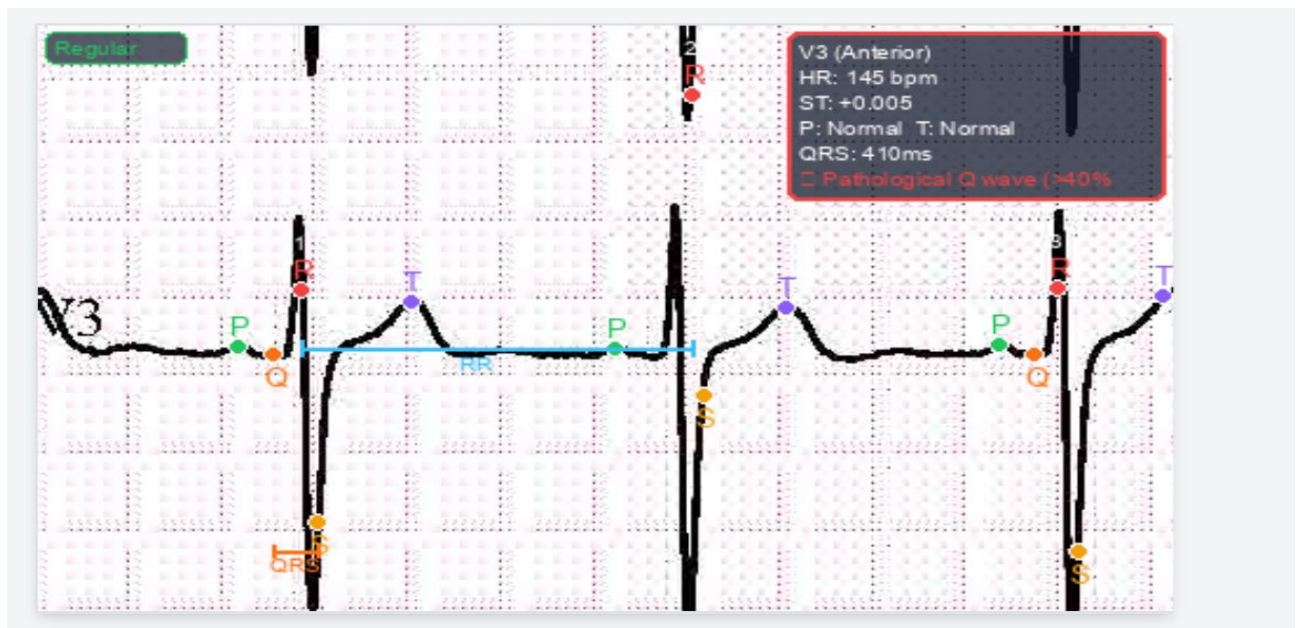


Fig. 5.17: Explainability analysis interface (single annotated ECG leads with PQRS)

Individual Classifier Votes

The ensemble combines 5 independent classifiers. Green border = agrees with final prediction.

SUPPORT VECTOR MACHINE	K-NEAREST NEIGHBORS	RANDOM FOREST	GAUSSIAN NAIVE BAYES	LOGISTIC REGRESSION
Normal 97.2%	Normal 60.0%	Normal 78.1%	Normal 100.0%	Normal 93.1%

Fig. 5.18: Explainability analysis interface (individual model votes)

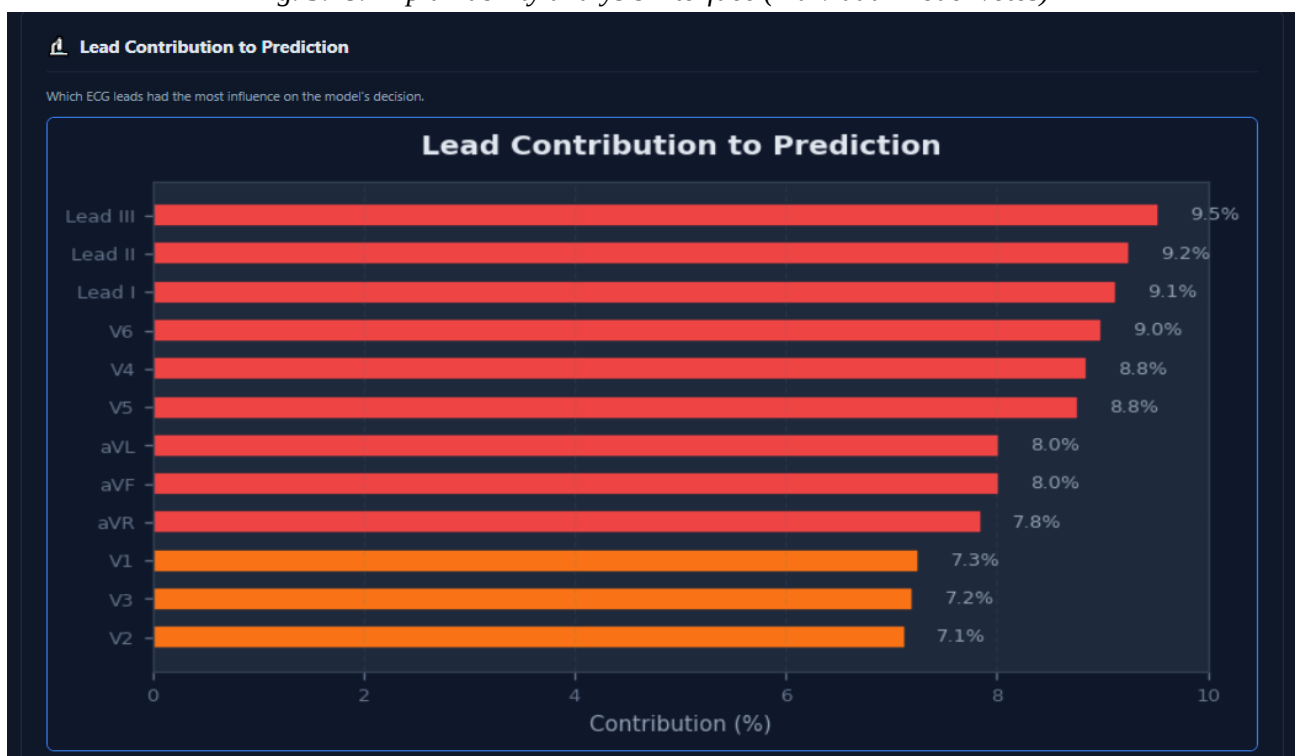


Fig. 5.19: Explainability analysis interface (lead-contribution information)

Clinical Interpretation Notes

AI-generated model-level observations.

ECG classification result: **Your ECG is Normal**

Strong consensus: All 5 ensemble classifiers unanimously agree.

Model confidence is moderate (79.5%). Clinical correlation advised.

Sinus tachycardia (HR: 108 bpm).

 **Disclaimer:** Computer-assisted analysis. All findings must be verified by a qualified physician.

Fig. 5.20: Explainability analysis interface (clinical interpretation notes)

5.5.15. Doctor Report Review Screen

The report-review screen, as shown in Fig. 5.21, enables doctors to carefully examine the AI-assisted ECG analysis before the report is finalized. It displays the diagnostic results and relevant clinical information generated by the system. Doctors can assess the accuracy and validity of the findings and make informed decisions regarding the report. The screen provides options to either approve or reject the generated report. Once approved, the report becomes available for patient access and download.

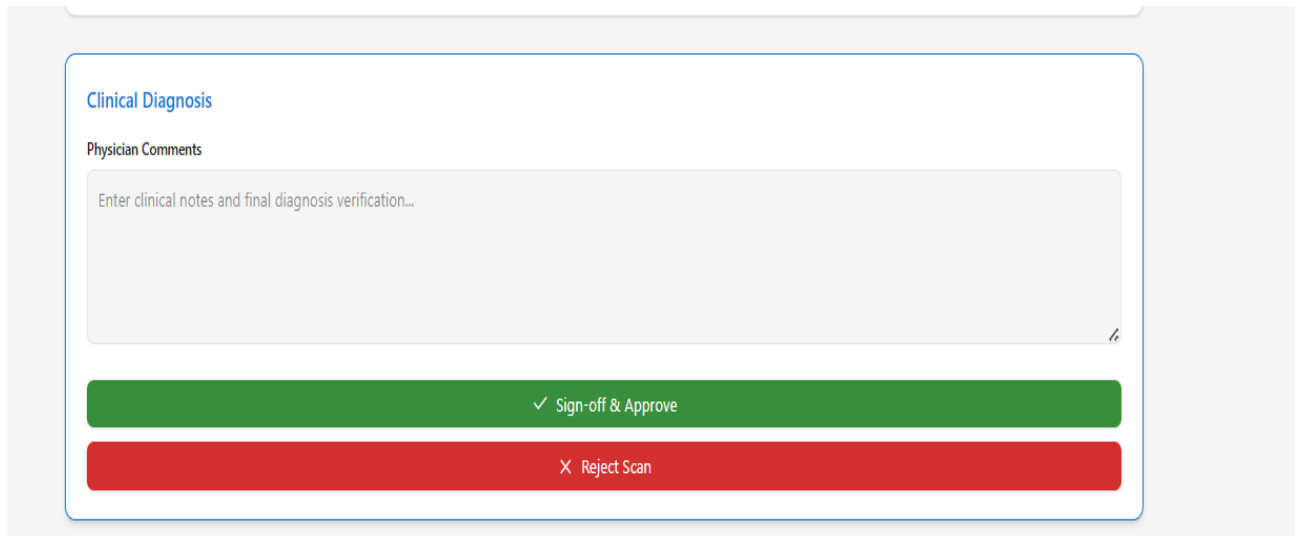


Fig. 5.21: Doctor Report review and approval interface

5.5.16. Patient Report View and Download Screen

The patient-report screen, as shown in Fig. 5.22, displays the approved ECG report generated by the system. It allows patients to review the diagnosis and related medical findings. All report details are presented in a clear and easy-to-read format. Users can access the finalized report once it has been approved by the doctor. A download option is provided to save the report file for future reference or sharing.

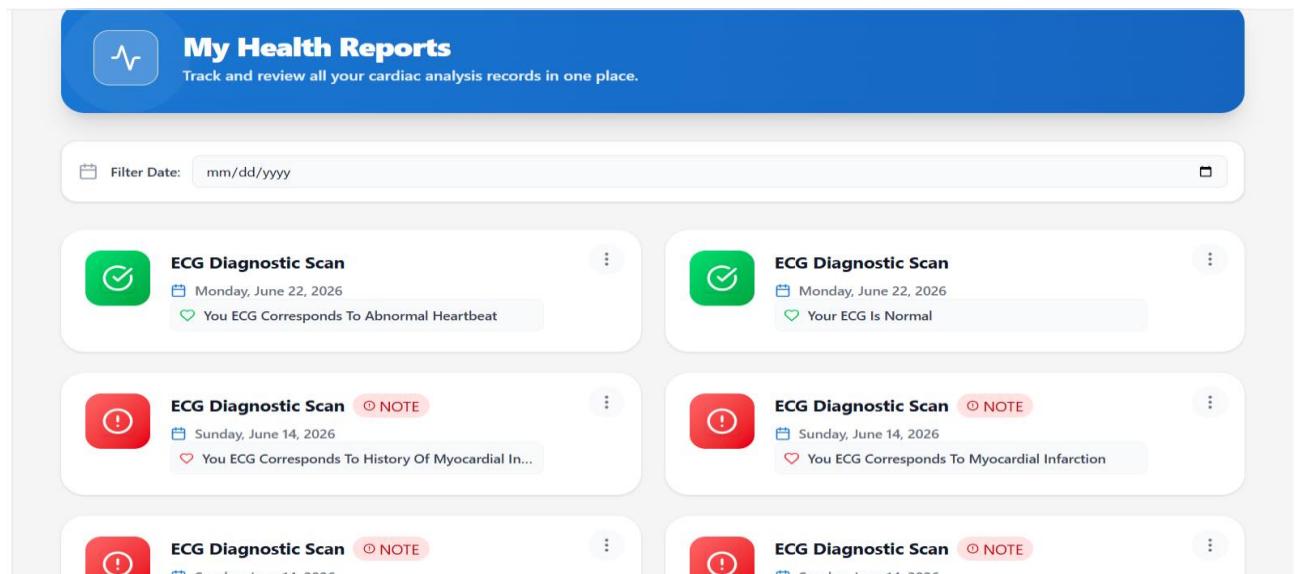


Fig. 5.22: Patient report list view and download interface

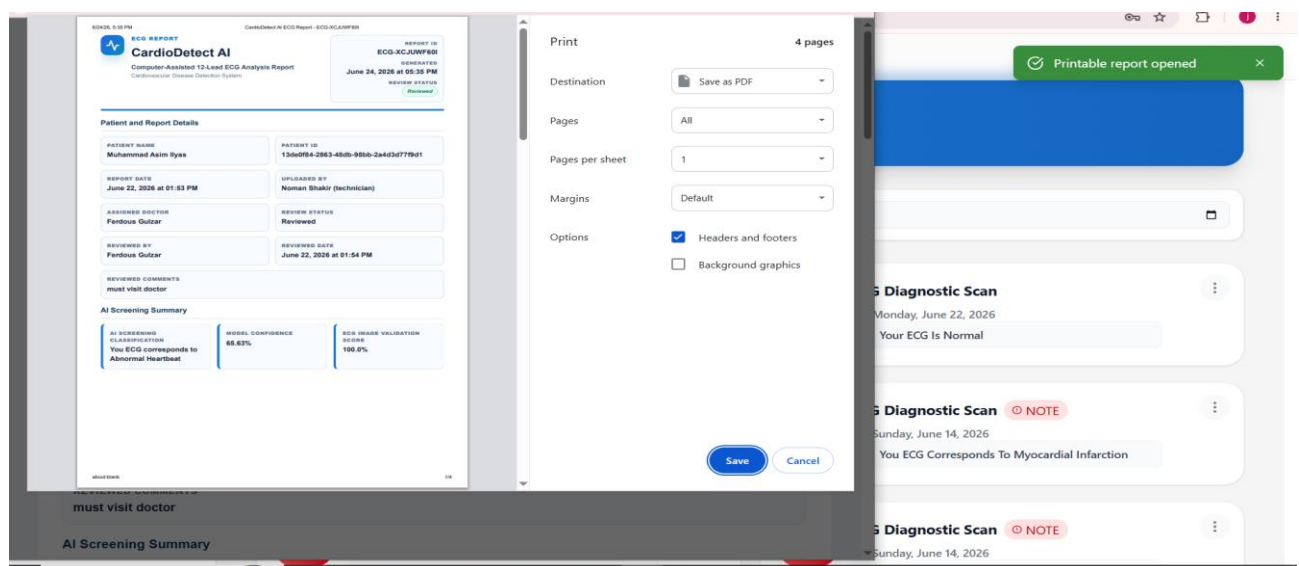


Fig. 5.23: Patient report view in pdf

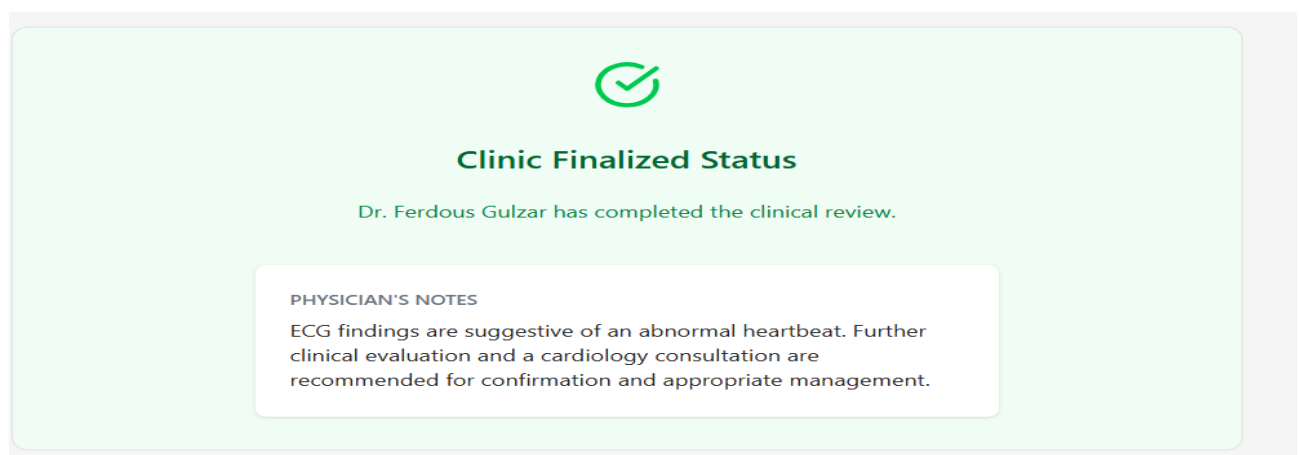


Fig. 5.24: Physician's Notes for Patient

5.5.17. Audit Logs and Staff Management Screen

The administrative management screen, as shown in Fig. 5.25, enables an authorized administrator to review audit events and manage staff accounts according to the role-based workflow.

Date	Time	Performed By	Action	Resource / Details	Status
6/8/2026	1:59:40 PM	Noman Shakir system: ef937d8-5567-4625-bf9c-ed927079923b	USER LOGIN	noman12@gmail.com (technician)	Successfully performed
6/8/2026	1:59:40 PM	Noman Shakir system: ef937d8-5567-4625-bf9c-ed927079923b	USER LOGIN	noman12@gmail.com (technician)	Successfully performed
6/8/2026	1:54:26 PM	Noman Shakir system: ef937d8-5567-4625-bf9c-ed927079923b	USER LOGIN	noman12@gmail.com (technician)	Successfully performed
6/8/2026	1:54:26 PM	Noman Shakir system: ef937d8-5567-4625-bf9c-ed927079923b	USER LOGIN	noman12@gmail.com (technician)	Successfully performed
6/6/2026	2:53:26 PM	Tech system: dev_user_1777288333625	USER LOGIN	tech@gmail.com (technician)	Successfully performed
6/6/2026	2:53:26 PM	Tech system: dev_user_1777288333625	USER LOGIN	tech@gmail.com (technician)	Successfully performed
6/6/2026	8:41:09 AM	Tech system: dev_user_1777288333625	USER LOGIN	tech@gmail.com (technician)	Successfully performed

System Status: All Systems Operational

Fig. 5.25: Audit logs interface

Name	Requested Role	Phone	Request Date	Actions
Snobar Gulzar snobar@gmail.com	Doctor 1111	+8326473264236	6/8/2026, 2:26:20 PM	Approve Reject

Approval Guidelines

- Verify that the user's credentials match their stated role
- Check organization details for healthcare professionals
- Doctors and Technicians require admin approval before accessing the system
- Once approved, users will be able to log in immediately
- Rejected users will not be able to access the system

Fig. 5.26: Staff-management interface

6. Testing and Evaluation

Testing and evaluation are essential activities in the software development lifecycle because they verify whether the implemented system satisfies its functional requirements, quality expectations, and clinical workflow constraints. This chapter evaluates the Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection through structured manual testing, unit-level verification, functional testing, integration testing, automated-test planning, and non-functional evaluation.

The testing scope covers the complete web-based workflow: account registration, role-based authentication, staff approval, patient selection, doctor assignment, ECG upload, ECG image validation, lead segmentation, preprocessing, feature generation, PCA transformation, cardiovascular disease prediction, explainability output, report review, report approval, report access, and audit monitoring. The tests have been written specifically for the implemented project and are aligned with the use cases and requirements defined in Chapter 3.

A key access-control rule is evaluated throughout this chapter. A patient may create an account and use the system directly after successful registration. In contrast, a doctor or technician may submit a registration request, but the account remains inactive until an administrator reviews and approves the request. The staff member must not be able to log in before approval. This approval mechanism protects medical records from unauthorized access and ensures that only verified clinical staff can use restricted modules.

6.1. Testing Objectives

The primary objective of testing is to confirm that the system behaves correctly under valid and invalid conditions while maintaining secure and consistent role-based workflows. The evaluation activities were designed around the following goals:

- i. Verify that patient registration and direct login work correctly after successful account creation.
- ii. Confirm that doctor and technician accounts remain pending after registration and cannot log in until approved by the administrator.
- iii. Validate that the administrator can approve or reject staff requests and that the selected action changes account access appropriately.
- iv. Ensure that authorized users can select patients and doctors, upload supported ECG images, and receive clear feedback for invalid inputs.
- v. Verify the internal ECG-processing flow, including structural similarity validation, twelve-lead segmentation, long-lead extraction, preprocessing, contour-based feature generation, PCA transformation, and four-class prediction.
- vi. Confirm that analysis progress and explainability information are presented in a structured manner for medical review.
- vii. Ensure that doctors can approve or reject reports and that patients can access only their own approved reports.

6.2. Test Environment and Evaluation Scope

The application was evaluated as an integrated web-based clinical decision-support system. The frontend provides role-specific dashboards, while the FastAPI backend coordinates ECG-processing operations. The processing pipeline receives an uploaded ECG image, performs preliminary validation, and streams intermediate states to the user interface. Table 6.1 summarizes the main components included in the evaluation scope.

Table 6.1: Components included in the testing scope

Component	Implementation Layer	Evaluation Scope
Frontend application	React with TypeScript	Role-based dashboards, registration forms, login screens, ECG upload interfaces, progress views, and analysis presentation.
Authentication and profiles	Supabase Authentication and database services	Patient self-registration, staff approval state, credential verification, session control, roles, and account status.
Backend API	FastAPI with Python	File reception, ECG validation, prediction orchestration, progress streaming, exception handling, and final response delivery.
Preliminary validation endpoint	POST /validate-ecg	Checks whether an uploaded image follows the expected ECG-sheet structure before full processing.
Prediction endpoint	POST /predict	Executes the ECG-processing pipeline and streams progress updates through server-sent events.
ECG-processing class	ECG class	Image reading, grayscale conversion, structural similarity checking, lead segmentation, preprocessing, contour extraction, scaling, CSV consolidation, PCA transformation, and classification.
Explainability component	Clinical explainability module	Class probabilities, model votes, lead-level observations, annotated visualizations, and interpretation support for clinical review.
User roles	Admin, Doctor, Technician, Patient	Tests role permissions, approval-dependent access, clinical workflow routing, and report visibility.

The manual test cases use non-sensitive sample accounts and representative ECG images. Patient accounts are tested as directly accessible accounts. Doctor and technician accounts are tested in both pending and approved states. ECG testing includes a valid image following the expected dataset layout, a non-ECG image, and a corrupted file. This combination verifies both successful processing and controlled rejection behavior.

The result column in the following tables records the current implementation-level verification outcome. Exact load-sensitive timings, concurrency limits, and production uptime values should be finalized through benchmark execution in the deployment environment because these values depend on hardware, network conditions, and hosting configuration.

6.3. Manual Testing

Manual testing involves executing test scenarios through the application interface without relying solely on automated scripts. It is useful for validating role-based workflows, user experience, error messages, and the visible behavior of the ECG-analysis process.

6.3.1. System Testing

System testing verifies the behavior of the complete application as a single integrated system. The test cases in Table 6.2 evaluate account access, staff approval, clinical workflows, ECG processing, explainability, reporting, and administrative controls.

Table 6.2: System testing

Test Case	Description	Expected Result	Actual Result	Status
Patient registration	Register a new patient using valid details.	Patient account is created and may proceed to login directly.	Patient profile is created and direct login is available.	Pass
Doctor registration request	Register a new doctor using valid staff details.	Request is stored with pending approval status.	Doctor request is submitted for administrator review.	Pass
Technician registration request	Register a new technician using valid staff details.	Request is stored with pending approval status.	Technician request is submitted for administrator review.	Pass
Staff login before approval	Attempt to log in using pending doctor or technician credentials.	System blocks access until administrator approval.	Pending staff account is not allowed to enter restricted modules.	Pass
Admin staff approval	Approve a pending doctor or technician request.	Account status becomes active and login becomes available.	Approved staff member can access the assigned dashboard.	Pass
Admin staff rejection	Reject a pending staff request.	Rejected account remains unable to log in.	Access is denied for the rejected request.	Pass
Role-based dashboard	Log in using each approved role.	System opens the dashboard permitted for the authenticated role.	Dashboard options change according to Admin, Doctor, Technician, or Patient role.	Pass
Valid ECG upload	Upload a supported ECG image linked to a patient.	System accepts the ECG image and starts analysis.	Valid ECG image proceeds to backend processing.	Pass
Invalid image rejection	Upload a non-ECG image.	System rejects the file and displays a clear validation message.	Invalid image is rejected before full prediction.	Pass
ECG-processing pipeline	Run analysis for an accepted ECG image.	System performs validation, lead separation, preprocessing, feature preparation, PCA transformation, and prediction.	Processing stages complete and the prediction response is generated.	Pass

Test Case	Description	Expected Result	Actual Result	Status
Progress streaming	Observe the interface during ECG analysis.	Progress messages and intermediate outputs appear while processing continues.	The interface receives staged updates from the backend.	Pass
Explainability presentation	Open the completed ECG analysis.	Prediction details and explainability outputs are displayed for clinical review.	The analysis view presents supporting visual and numerical information.	Pass
Doctor report review	Open a pending report as an approved doctor.	Doctor can review the ECG result and select an appropriate decision.	Doctor review workflow is available.	Pass
Report approval	Approve a reviewed report as doctor.	Report status changes to approved and becomes accessible to the patient.	Approved report is available for patient viewing and download.	Pass
Audit-log monitoring	Open audit logs as administrator.	Administrator can review recorded system activities.	Audit-log screen is accessible to the administrator role.	Pass
Logout	Log out from an authenticated account.	Session terminates and protected pages require re-authentication.	User is redirected to the login screen.	Pass

The system-testing results demonstrate that the main modules operate together in a controlled workflow. The staff-approval rule is particularly important: a doctor or technician cannot bypass the administrator review stage, while patients retain a straightforward self-registration path.

6.3.2. Unit Testing

Unit testing focuses on individual functions and small workflow units. Tables 6.3–6.6 separate account-related behavior from ECG-processing behavior so that failures can be traced to a specific component.

Table 6.3: Registration and staff-approval unit test cases

No.	Test Case / Test Script	Attribute and Value	Expected Result	Result
1	Register a patient using valid details.	Role: Patient; Email: patient@gmail.com	Patient account is created and can proceed to login.	Pass
2	Register a patient using an existing email.	Email: patient@gmail.com	System rejects duplicate registration and displays an informative message.	Pass
3	Register using an invalid email format.	Email: patient.test@	System blocks submission and requests a valid email address.	Pass
4	Submit a doctor registration request.	Role: Doctor; Email: doctor@gmail.com	Doctor account is stored with pending approval status.	Pass
5	Submit a technician registration request.	Role: Technician; Email: technician@gmail.com	Technician account is stored with pending approval status.	Pass

No.	Test Case / Test Script	Attribute and Value	Expected Result	Result
6	Approve a pending doctor account as administrator.	Action: Approve doctor request	Account becomes active and may log in.	Pass
7	Reject a pending technician account as administrator.	Action: Reject technician request	Account remains inaccessible.	Pass

Table 6.4: Login and session unit test cases

No.	Test Case / Test Script	Attribute and Value	Expected Result	Result
1	Verify patient login with valid credentials.	patient@gmail.com / Valid password	Patient is authenticated and redirected to the patient dashboard.	Pass
2	Verify approved doctor login.	doctor@gmail.com.co m / Valid password	Approved doctor is authenticated and redirected to the doctor dashboard.	Pass
3	Verify pending staff login.	Pending doctor or technician account	System denies login and informs the user that administrator approval is required.	Pass
4	Verify login using an incorrect password.	Valid email / Incorrect password	System rejects authentication and displays an invalid-credentials message.	Pass
5	Verify login using an unregistered email.	unknown@gmail.com	System rejects authentication and displays an appropriate message.	Pass
6	Verify login with a blank email field.	Email: blank	System prompts the user to enter a valid email.	Pass
7	Verify logout and session termination.	Action: Logout	Session is cleared and protected pages require login again.	Pass

Table 6.5: ECG-image validation unit test cases

No.	Test Case / Test Script	Attribute and Value	Expected Result	Result
1	Read a valid RGB ECG image.	getImage(image)	Image is loaded without error.	Pass
2	Read an ECG image containing an alpha channel.	RGBA image	Alpha channel is removed and RGB data is retained.	Pass
3	Convert an accepted ECG image to grayscale.	GrayImage(image)	Grayscale image is resized to 1572 x 2213 pixels.	Pass
4	Validate a representative ECG image.	POST /validate-ecg with valid ECG	Response contains is_ecg = true when maximum SSIM score is at least 0.70.	Pass
5	Validate a non-ECG image.	POST /validate-ecg with unrelated image	Response contains is_ecg = false and a rejection message.	Pass
6	Validate a corrupted image file.	Invalid or unreadable image bytes	System returns a controlled invalid-format message instead of crashing.	Pass

Table 6.6: ECG-processing and prediction unit test cases

No.	Test Case / Test Script	Attribute and Value	Expected Result	Result
1	Divide a validated ECG sheet.	DividingLeads(image)	Method returns twelve standard leads and one long-lead region.	Pass
2	Preprocess the separated leads.	PreprocessingLeads(leads)	Lead images are converted to binary form using smoothing and Otsu thresholding.	Pass
3	Extract contours from standard leads.	SignalExtraction_Scaling(leads)	A representative contour is extracted and resized for each standard lead.	Pass
4	Normalize lead-level signal values.	MinMaxScaler	Extracted values are normalized before feature consolidation.	Pass
5	Store lead-level values.	Scaled_1DLead_n.csv	CSV files are generated for the twelve standard leads.	Pass
6	Combine one-dimensional lead signals.	CombineConvert1Dsignal()	Lead-level CSV values are consolidated into a model-input dataframe.	Pass
7	Reduce feature dimensionality.	DimensionalReduction(dataframe)	The trained PCA model transforms the combined feature representation.	Pass
8	Generate a disease prediction.	ModelLoad_predict(pca_features)	Classifier returns one supported cardiovascular disease category.	Pass
9	Create final API response.	POST /predict	Response includes success status, prediction, similarity score, confidence, and explainability payload.	Pass

6.3.3. Functional Testing

Functional testing verifies that each user-visible module satisfies the requirements defined in Chapter 3. Table 6.7 maps the implemented behavior to the major functional requirements and expected outcomes.

Table 6.7: Functional testing of major requirements

No.	Requirement	Test Case / Test Script	Attribute and Value	Expected Result	Result
1	FR-1	Register a new patient account.	Valid patient profile and credentials	Account is created and patient can proceed to login directly.	Pass
2	FR-1 / FR-10	Register a doctor account.	Valid doctor profile and credentials	Request is submitted and remains pending until administrator approval.	Pass
3	FR-1 / FR-10	Register a technician account.	Valid technician profile and credentials	Request is submitted and remains pending until administrator approval.	Pass
4	FR-2	Log in using a pending doctor or technician account.	Pending account credentials	Access is denied until approval is granted.	Pass

No .	Requirement	Test Case / Test Script	Attribute and Value	Expected Result	Result
5	FR-10	Approve a pending staff request.	Admin selects Approve	Staff account becomes active.	Pass
6	FR-10	Reject a pending staff request.	Admin selects Reject	Rejected staff account remains inactive.	Pass
7	FR-2	Log in using valid active credentials.	Approved user credentials	User is redirected to the appropriate role-based dashboard.	Pass
8	FR-3	Log out from an active session.	Action: Logout	Session is terminated securely.	Pass
9	FR-12	Search and select a patient.	Patient name or identifier	Selected patient becomes the active context for subsequent actions.	Pass
10	FR-13	Search and select an approved doctor.	Doctor name or specialty	Verified doctor is linked to the ECG workflow.	Pass
11	FR-5	Upload a valid JPG or PNG ECG image.	Supported ECG image	File is accepted and forwarded for AI processing.	Pass
12	FR-5	Upload a non-ECG or corrupted file.	Invalid image	System rejects the input and displays a clear message.	Pass
13	FR-4	View completed ECG analysis.	Processed ECG record	Doctor can view prediction, confidence, supporting images, and explainability information.	Pass
14	FR-6	Approve an ECG report.	Doctor decision: Approve	Status changes to approved and patient access becomes available.	Pass
15	FR-6	Reject or request changes for a report.	Doctor decision: Reject / Request changes	Status is updated and the workflow returns for revision.	Pass
16	FR-7	View patient details.	Authorized doctor or technician	Relevant demographics, ECG history, and reports are displayed.	Pass
17	FR-8 / FR-11	View and download an approved report.	Patient opens own approved report	Report is available for viewing and download.	Pass
18	FR-9	View and filter audit logs.	Administrator role	Recorded system activities are visible to the administrator.	Pass

6.3.4. Integration Testing

Integration testing, as shown in Table 6.8, validates the communication and interaction between the frontend, authentication layer, database services, backend API, ECG-processing class, prediction model, explainability component, and report workflow. The test cases verify that data is transmitted accurately across all integrated modules, ensuring seamless execution of the ECG analysis process from user input to report generation. These tests also confirm that the integrated components exchange information correctly, maintain the required clinical context, and operate together reliably to support the overall functionality of the proposed cardiovascular disease detection system.

Table 6.8: Integration test cases

No.	Integration Scenario	Test Case / Test Script	Expected Result	Result
1	Patient registration to direct login	Register a patient and log in using the same credentials.	Patient profile is stored and the patient dashboard becomes accessible.	Pass
2	Staff registration to pending state	Register a doctor or technician.	Staff profile is stored as pending and restricted login remains blocked.	Pass
3	Admin approval to staff login	Approve the pending staff request and repeat login.	Approved account receives the correct role-based dashboard.	Pass
4	Patient selection to ECG upload	Technician selects a patient before uploading an ECG image.	Uploaded ECG record is associated with the selected patient.	Pass
5	Doctor selection to workflow assignment	Technician selects an approved doctor.	The report is routed to the selected verified doctor for review.	Pass
6	Frontend upload to validation endpoint	Upload an ECG image through the interface.	Frontend sends file data and receives the preliminary validation result.	Pass
7	Validation to processing pipeline	Use an ECG image with SSIM score greater than or equal to 0.70.	Accepted image proceeds to full analysis.	Pass
8	Invalid input to controlled termination	Use a non-ECG image or corrupted file.	Pipeline stops early and returns an understandable error message.	Pass
9	Processing pipeline to streamed UI progress	Run POST /predict with a valid ECG image.	Backend streams intermediate progress messages and images to the interface.	Pass
10	Feature extraction to classifier	Complete segmentation and preprocessing.	Combined feature dataframe is transformed by PCA and processed by the trained classifier.	Pass
11	Prediction to explainability	Complete classification successfully.	Explainability payload is generated and returned with the final response.	Pass
12	Doctor approval to patient access	Doctor approves a completed ECG report.	Patient can view and download the approved report.	Pass
13	Report rejection to access restriction	Doctor rejects or requests changes for a report.	Unapproved report does not become a final patient-downloadable record.	Pass
14	Restricted route protection	Attempt to open a privileged module using an unauthorized role.	System denies access and maintains role boundaries.	Pass
15	Logout to protected-route lockout	Log out and revisit a protected route.	System redirects to login and requires re-authentication.	Pass

6.4. Automated Testing

Automated testing is valuable for repeatable regression checking, particularly for backend endpoints and ECG-processing functions. The current chapter documents a practical automated-test plan for the final integrated build. The plan avoids replacing manual clinical-workflow verification; instead, it complements manual testing by checking repetitive API and processing behaviors consistently.

6.4.1. Tools Used for Automated Testing

This section describes the tools used to automate repetitive test cases, validate key system modules, and improve the consistency of the testing process. Table 6.9 presents the tools used for automated testing and regression verification.

Table 6.9: Tools used for automated testing and regression verification

Tool Name	Tool Description	Applied On	Evaluation Use
Pytest	Python testing framework used to organize unit and regression tests.	ECG class methods, validation logic, PCA transformation, classifier response format, and utility functions.	Suitable for repeatable backend verification.
FastAPI TestClient	Test client for invoking FastAPI endpoints without manually operating the browser interface.	POST /validate-ecg and POST /predict endpoint behavior, status handling, and response structure.	Suitable for API-level automated checks.
Postman Collection Runner	API regression tool used to store endpoint requests and rerun them systematically.	Positive and negative file-upload scenarios, endpoint smoke tests, and error-message verification.	Suitable for repeatable integration smoke tests.
Browser Developer Tools	Browser-based network and performance inspection utilities.	Frontend requests, progress-event delivery, page responsiveness, and client-side errors.	Suitable for final UI and performance review.
Supabase Dashboard Logs	Administrative view for checking stored profiles, account status, and database activity.	Patient activation, staff pending status, administrator approval, role assignment, and record linkage.	Suitable for data-layer verification.

6.4.2. Automated Regression Test Plan

This section presents the repeatable test cases executed after system updates to ensure that previously implemented features continue to function correctly. Table 6.10 presents the automated regression test plan.

Table 6.10: Automated regression test plan

Test ID	Automated Test Case	Target	Expected Assertion
AT-01	Validate a supported ECG image.	POST /validate-ecg	Response is successful, is_ecg is true, and confidence is greater than or equal to 0.70.
AT-02	Reject a non-ECG image.	POST /validate-ecg	Response is successful but is_ecg is false with an explanatory message.
AT-03	Handle corrupted image data safely.	POST /validate-ecg	Controlled invalid-format response is returned without backend crash.

Test ID	Automated Test Case	Target	Expected Assertion
AT-04	Complete prediction for a valid ECG image.	POST /predict	Stream reaches the final result event and includes prediction, similarity score, confidence, and explainability.
AT-05	Stop prediction for an invalid image.	POST /predict	Stream returns a validation error before feature extraction and classification.
AT-06	Verify lead segmentation output count.	DividingLeads()	Method returns thirteen ordered image regions: twelve standard leads and one long lead.
AT-07	Verify PCA transformation output.	DimensionalReduction()	Combined feature dataframe is transformed without shape or compatibility failure.
AT-08	Verify classifier output categories.	ModelLoad_predict()	Output maps to one of the four supported classes.
AT-09	Verify pending-staff access restriction.	Authentication workflow	Doctor or technician cannot access protected modules until approved.
AT-10	Verify approval-state transition.	Admin staff-management workflow	Approved account changes from pending to active and gains permitted access.

6.5. Evaluation against Non-Functional Requirements

The system was also evaluated against the non-functional requirements defined in Chapter 3. Some criteria can be checked directly through the integrated application, while deployment-dependent criteria require final benchmarking in the intended hosting environment. Table 6.11 records the appropriate evaluation method and current status for each major requirement.

Table 6.11: Evaluation against non-functional requirements

ID	Requirement	Evaluation Method	Current Observation	Status
USE-1	Technician uploads an ECG within three clicks.	Walk through the technician upload interface.	Verified through UI workflow	Pass
USE-2	Doctor opens new ECG results from the dashboard in a single click.	Open a pending result from the doctor dashboard.	Verified through UI workflow	Pass
USE-3	Patient accesses an approved report in two steps.	Open patient dashboard and select an approved report.	Verified through UI workflow	Pass
USE-4	Clear messages appear for invalid uploads.	Upload a non-ECG or corrupted file.	Validation message is returned	Pass
PER-1	Web pages load in under three seconds on a normal connection.	Measure browser load time under controlled network conditions.	Deployment-dependent benchmark	Pending Benchmark
PER-2	ECG preprocessing completes within five seconds.	Record backend timing for validation, segmentation, and preprocessing.	Hardware-dependent benchmark	Pending Benchmark

ID	Requirement	Evaluation Method	Current Observation	Status
PER-3	Prediction with explainability completes in approximately ten seconds.	Measure timestamps from first to final streamed event.	Hardware-dependent benchmark	Pending Benchmark
PER-4	Report generation completes within five seconds.	Measure report-generation time after approval.	Deployment-dependent benchmark	Pending Benchmark
PER-5	System supports up to fifty active users without slowdown.	Run a controlled load test.	Requires concurrency benchmark	Pending Load Test
REL-1	System maintains at least 95% uptime during operating hours.	Collect hosting-monitoring data over time.	Requires deployed monitoring period	Production Evaluation
REL-2	ECG processing retries automatically after an initial failure.	Trigger a controlled processing failure and observe retry behavior.	Requires final implementation verification	Review Required
REL-3	Important actions are logged with timestamp and user ID.	Review audit logs after authentication and report actions.	Verified through admin workflow	Pass
SEC-1	Role-based access restricts modules appropriately.	Attempt privileged actions using unauthorized roles.	Access-control tests passed	Pass
SEC-2	Data transmission occurs over HTTPS.	Inspect deployed network configuration.	Hosting configuration dependent	Deployment Check
MAINT-1	System uses a modular structure.	Review frontend, backend, ECG-processing, and explainability modules.	Module separation present	Pass
SCA-1	AI processing supports multiple workers.	Review backend executor configuration and run load checks.	Worker pool config.d; load test pending	Partially Verified
PORT-1	System operates on Windows.	Run application in the development environment.	Windows execution confirmed	Pass
PORT-2	System supports major browsers.	Repeat UI smoke tests in Chrome, Firefox, Edge, and Safari.	Cross-browser validation required	Pending Browser Test
DATA-1	Each ECG record remains linked to the correct patient.	Upload ECG after selecting a patient and inspect the stored context.	Verified through integration workflow	Pass
DATA-2	Prediction metadata is saved for each result.	Inspect stored result fields for confidence, time, and model information.	Requires final database verification	Review Required

The evaluation distinguishes completed functional verification from deployment-dependent benchmarking. Functional testing validates core system features, while performance, browser compatibility, HTTPS configuration, concurrency, and production uptime should be evaluated in the final deployment environment before system release.

7. Conclusion and Future Work

This chapter concludes the Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection by summarizing the major outcomes of the project and identifying realistic directions for its continued development. The work has produced an ECG-based web application that combines structured clinical workflows with automated image processing, machine-learning inference, and explainable result presentation. The chapter also discusses current limitations so that future improvements can be planned without overstating the capabilities of the present system.

The proposed platform is designed as a clinical decision-support tool rather than a replacement for professional medical judgement [17]. It focuses exclusively on ECG-based cardiovascular disease analysis and supports a controlled workflow in which patient records, uploaded ECG images, AI-generated findings, doctor review decisions, and approved reports remain connected within a role-based system.

7.1. Conclusion

The Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection was developed to address the need for a more organized, interpretable, and efficient ECG-analysis workflow. The system brings together web-based access, ECG image preprocessing, feature generation, classification, explainability, report handling, and role-specific controls in a single platform. This integration reduces the fragmentation commonly observed when ECG analysis is treated only as an isolated prediction task.

The implemented application supports four primary users: Admin, Doctor, Technician, and Patient. A patient can create an account and log in directly after successful registration. In contrast, a doctor or technician submits a registration request and cannot access restricted modules until the administrator reviews and approves the account. This approval-dependent workflow provides an additional layer of control for clinical roles. The administrator manages staff requests and audit-related activities, the technician uploads ECG files and links cases with the relevant patient and doctor, the doctor reviews AI-assisted findings and approves or rejects reports, and the patient views approved results.

At the technical level, the ECG-analysis pipeline accepts an uploaded ECG image, validates its structure through a similarity check, converts it into a standardized representation, and segments the sheet into twelve standard leads together with one long-lead region. The standard leads are processed using grayscale conversion, Gaussian smoothing, Otsu thresholding, and binary-image generation. Contour extraction is then used to derive normalized one-dimensional ECG features. The generated lead-wise values are combined into a single vector, reduced through Principal Component Analysis (PCA), and submitted to the trained classification model. The backend also streams intermediate processing stages to the frontend so that users can observe the progress of the analysis.

The deployed classifier recognizes four output categories: Abnormal Heartbeat, Myocardial Infarction, Normal ECG, and History of Myocardial Infarction. After classification, the explainability component presents class probabilities, individual model

votes, lead-wise observations, annotated ECG images, lead-contribution information, axis estimates, and computer-assisted interpretation notes. These outputs improve transparency and support the doctor during report review.

7.2. Summary of Experimental Model Performance

The experimental phase compared multiple machine-learning and deep-learning configurations to examine the effect of feature representation and dimensionality reduction. The original one-dimensional input contained 3,060 values per ECG record, calculated from twelve leads with 255 values per lead. PCA reduced this representation to 400 principal components while retaining the dominant information required for classification. Table 7.1 summarizes the principal results reported during the model-testing phase.

Table 7.1: Summary of experimental model performance

Experiment	Accuracy	Precision	Recall	F1-score	Observation
SVM without PCA	0.9052	0.91	0.89	0.90	Best individual non-PCA ML result
SVM with PCA	0.90	0.90	0.90	0.90	Strong PCA-based individual model
Soft-voting ensemble	0.9347	0.93	0.92	0.93	Macro-average ensemble result
LSTM on raw 1D signals	0.828	0.82	0.83	0.81	Sequence-model baseline
LSTM on PCA representation	0.849	0.85	0.85	0.82	Improved LSTM configuration
ResNet50 on whole ECG images	0.9301	0.93	0.93	0.93	Highest isolated experimental score

The soft-voting ensemble combines the probability outputs of Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Random Forest, Gaussian Naive Bayes, and Logistic Regression models. Grid-search optimization and five-fold cross-validation were used during the experimental setup. The ensemble achieved an accuracy of approximately 93.47%, with a macro-average precision of 0.92, macro-average recall of 0.91, and macro-average F1-score of 0.91. The whole-ECG ResNet50 experiment produced the highest isolated score of approximately 93.01%. These results provide a strong basis for further evaluation of decision-level fusion strategies in the production version of the system.

The current web workflow described in the implementation chapter uses the serialized classifier after PCA transformation. The experimental deep-learning results should therefore be interpreted as evaluated model candidates and as a foundation for future production-level fusion, rather than as evidence that every tested model has already been integrated into the live inference path.

7.3. Project Contributions

The project makes the following practical and technical contributions:

- It provides an end-to-end web-based ECG-analysis workflow that connects user management, ECG upload, preprocessing, prediction, explainability, doctor review, and approved report access.

- ii. It implements role-based access for Admin, Doctor, Technician, and Patient users while enforcing administrator approval before doctors and technicians can log in.
- iii. It converts ECG sheets into structured numerical representations through lead separation, image preprocessing, contour extraction, normalization, feature combination, and PCA transformation.
- iv. It evaluates multiple machine-learning and deep-learning approaches and establishes a quantitative basis for selecting robust classification strategies.
- v. It presents explainability information alongside predictions so that the result is easier to inspect during clinical review.
- vi. It retains the role of the doctor as the final reviewer by requiring approval or rejection of AI-assisted reports before patient access.

7.4. Current Limitations

Although the developed system provides a functional and extensible foundation, several limitations remain and should be acknowledged clearly. These limitations define the boundary of the present work and guide future enhancements.

- i. The current pipeline is designed for ECG images that follow the expected sheet structure. Images captured from different devices, layouts, resolutions, or scanning conditions may require additional preprocessing and adaptive segmentation.
- ii. The available evaluation is based on the selected ECG dataset and project test environment. Independent clinical evaluation using ECG records from additional hospitals and devices is required before real-world medical deployment.
- iii. The platform provides decision support only. It does not replace cardiologist interpretation and does not provide treatment recommendations, medication advice, or emergency-response functionality.
- iv. The initial release is limited to ECG-based analysis. Non-ECG medical imaging modalities, such as MRI, CT, and X-ray, remain outside the project scope.
- v. Production-level performance measurements, including hosted response time, uptime, cross-browser behavior, HTTPS configuration, and concurrency capacity, require final deployment benchmarking.
- vi. The experimental deep-learning configurations and the deployed PCA-based classifier require further controlled integration studies before a final production-level fusion strategy is selected.

7.5. Future Work

The current implementation establishes a strong basis for continued improvement. Future work should extend the system carefully while preserving its ECG-focused clinical decision-support purpose. The most relevant enhancements are described below.

7.5.1. Broader Dataset Evaluation and Clinical Validation

Future evaluation should include ECG data collected from additional hospitals, devices, and patient groups. Independent test data would help determine whether the model maintains reliable performance beyond the original dataset. Evaluation should also report class-wise error patterns, calibration behavior, and performance under different image-quality conditions. Cardiologist review of a representative sample of outputs would further strengthen the clinical relevance of the system [17].

7.5.2. Production-Level Decision Fusion

The next model-development stage should evaluate the controlled integration of validated one-dimensional and two-dimensional ECG representations. The soft-voting ensemble, the whole-ECG CNN experiment, and other ECG-derived representations can be compared through decision-level fusion. The final production strategy should be selected only after measuring accuracy, robustness, explainability quality, inference time, and generalization on independent test data.

7.5.3. Robust ECG Input Handling

The ECG-upload workflow can be extended to support a wider range of valid sheet formats and image-acquisition conditions. Future improvements may include adaptive lead-region detection, image-quality assessment, clearer feedback for low-quality uploads, and improved handling of scanned or mobile-captured ECG sheets. These changes would make the pipeline more practical without expanding beyond ECG-based analysis.

7.5.4. Explainability Review and Report Enhancement

The explainability component can be refined through structured review with medical experts. Future versions should evaluate whether lead-wise observations, annotated ECG images, contribution values, and interpretation notes remain clinically meaningful across different cases. The generated report can also be improved by adding quality warnings, model-version details, prediction timestamps, and a concise explanation of the evidence considered by the system.

7.5.5. Secure Deployment and Healthcare Interoperability

A production release should strengthen security and operational controls through HTTPS enforcement, deployment monitoring, backup procedures, audit-log review, and periodic access verification. Interoperability with hospital information systems can be explored in a later release through standards-based healthcare data exchange, such as HL7 FHIR [18]. This enhancement should be introduced only after privacy, authorization, and data-governance requirements have been reviewed carefully [19].

7.5.6. Model Monitoring and Maintenance

A deployed clinical decision-support system should record the model version, prediction timestamp, confidence value, and processing status for each ECG analysis. Future versions can use these records to monitor model behavior over time, detect changes in incoming data, review incorrect predictions, and schedule controlled retraining when sufficient verified data becomes available. This lifecycle approach

would improve maintainability and accountability.

7.5.7. User Experience and Operational Improvements

The web interface can be enhanced through clearer status notifications, improved filtering of patient records, streamlined report-search options, responsive layouts, and accessibility improvements. Notifications for staff-account approval, ECG-processing completion, doctor-review status, and report availability would further improve coordination among users. Load testing and cross-browser validation should also be completed before final deployment.

Table 7.2 presents the recommended future-work roadmap for the proposed system. It outlines potential enhancements, their expected benefits, and the corresponding implementation priorities to guide future development and deployment.

Table 7.2: Recommended future-work roadmap

Area	Recommended Enhancement	Expected Benefit	Priority
Clinical evaluation	Test on independent hospital and device data	Measure generalization and clinical reliability	High
Model development	Evaluate production-level ECG decision fusion	Improve robustness while preserving ECG-only scope	High
Input handling	Support more ECG layouts and image-quality checks	Reduce rejected or incorrectly processed uploads	High
Explainability	Conduct cardiologist review of explanations and reports	Improve transparency and clinical usefulness	High
Deployment	Complete HTTPS, load, browser, and uptime evaluation	Prepare the platform for controlled hosting	Medium
Interoperability	Explore standards-based EHR integration	Enable structured exchange with hospital systems	Medium
Maintenance	Add model monitoring and controlled retraining workflow	Support long-term reliability and accountability	Medium

7.6. Final Remarks

The Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection demonstrates how ECG preprocessing, multiple-model experimentation, explainable AI, and web-based clinical workflows can be brought together within a single software-engineering project. The verified PCA-assisted soft-voting ensemble achieved accuracy of 93.55% with a serialized classifier size of approximately 7.72 MB. The system provides a practical foundation for AI-assisted ECG analysis while maintaining the importance of medical oversight. Its modular architecture, role-based access, and explainability-oriented presentation make it suitable for continued refinement through broader validation, secure deployment, and carefully evaluated ECG-focused enhancements.

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Hybrid Multimodal Fusion Framework for Cardiovascular Disease Detection

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